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Dual-channel hierarchical interactive learning for the prediction of Protein-Ligand binding affinity

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ABSTRACT

Protein-ligand binding affinity (PLBA) is a crucial metric in drug screening for identifying potential candidate compounds. In recent years, deep learning-based methods have used representation learning to model interactions within protein-ligand complexes, demonstrating great promise in affinity prediction tasks. Existing studies have considered both intramolecular (covalent) and intermolecular (non-covalent) interactions to some extent. However, these interactions are often treated as independent features, lacking explicit hierarchical dependency modeling, which may lead to insufficient representation of interaction information and ultimately limit the accuracy of affinity predictions. To address this issue, we propose a novel approach—Dual-channel Hierarchical Interactive Learning (DHIL)—to achieve a more comprehensive modeling of protein-ligand interactions. DHIL employs a dual-channel encoding structure to simultaneously learn intramolecular and intermolecular interactions, ensuring the completeness of interaction features. Additionally, we design a hierarchical interactive learning paradigm to facilitate information exchange between these two interaction types at multiple levels, promoting their collaborative modeling. This mechanism mimics the local-to-global working principles of biological systems, enabling a more detailed and holistic representation of protein-ligand interactions. We conduct extensive and comprehensive experiments on a diverse set of benchmark datasets, rigorously evaluating the effectiveness of DHIL. The results demonstrate that DHIL significantly improves PLBA prediction accuracy, outperforming existing methods and further validating its potential in drug discovery and screening tasks.
Nevertheless, the proposed framework introduces notable computational overhead due to multi-scale graph construction and cross-level message passing. It also exhibits sensitivity to the quality of input 3D binding conformations, which may affect its robustness in practical applications. These limitations suggest future directions for improving model efficiency and generalizability. To facilitate reproducibility and further research, the complete source code of DHIL has been released at: https://github.com/WZY-0814/DHIL.

1. Introduction

Protein-Ligand Binding Affinity (PLBA) refers to the binding or interaction strength between ligands and target proteins in biological systems mediated by molecular microforces, serving as an important indicator for discovering desired candidate drugs (Kola & Landis, 2004). Here, ligands, including small molecules and biologics (typically referring to drug candidates), can act as agonists or inhibitors, binding to target proteins to form protein-ligand complexes, thus regulating signal transduction, metabolic pathways, and cell cycles in biological systems (Chen & Zhi, 2001). Traditionally, the binding affinity between a protein and a ligand is measured experimentally using parameters such as the dissociation constant (K_d), inhibition constant (K_i), and half maximal inhibitory concentration (IC_{50}) (He et al., 2017; Kairys et al., 2019). Nevertheless, those assays are labor-intensive and time-consuming. In recent times, advancements in computer-aided simulation techniques and data-driven algorithms have enabled the prediction of binding affinities during the early stages of drug discovery, which assists researchers in sequencing drug candidates and selecting the most promising ones for further testing, thereby expediting the process of virtual high-throughput screening (Lyne, 2002; Zhao et al., 2020).

Traditional computer-assisted approaches for PLBA predictions, such as molecular docking and physics-based scoring strategies, rely on empirical or semi-empirical interaction models constructed using structural domain knowledge (Beddell et al., 1976; Su et al., 2018b; Trott & Olson, 2010). While these methods provide strong interpretability, they

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often encounter challenges related to high computational cost, limited prediction accuracy, and insufficient generalizability when applied to large-scale compound libraries. In recent years, data-driven deep learning approaches exhibit unprecedented advantages by learning regression mappings between molecular descriptors and binding affinity metrics, achieving substantial improvements in both predictive accuracy and computational efficiency. Among these, Graph Neural Networks (GNNs) demonstrate exceptional potential by representing proteins and ligands as two- or three-dimensional graph structures and capturing structural and chemical contextual information through message passing mechanisms (Chu et al., 2022; Hua et al., 2023; Li et al., 2021). For such models, the ability to comprehensively characterize the complex and diverse interaction features between proteins and ligands remains a critical factor in achieving accurate PLBA predictions.

Against this background, emerging studies begin to incorporate both intramolecular (covalent) and intermolecular (non-covalent) interactions into unified modeling frameworks to further enhance prediction performance (Jiang et al., 2021; Li et al., 2020; Wang et al., 2023). Intramolecular interactions, mediated by covalent bonds, typically involve conformational adjustments during the binding process and play a key role in maintaining molecular stability and enabling structural optimization (Rohn, 2014). In contrast, intermolecular interactions—such as hydrogen bonding, van der Waals forces, and hydrophobic effects—serve as essential forces for stabilizing binding states and facilitating molecular recognition (Černỳ & Hobza, 2007). For example, the IGN model (Jiang et al., 2021) employs two sequential graph convolutional modules to separately model intramolecular and intermolecular interactions and introduces intermolecular features into downstream tasks to enhance performance. **However, most existing methods treat the two interaction types as independent features, without explicitly modeling their hierarchical hierarchical collaborations. This limitation constrains the expressive capacity and predictive power of these models in complex interaction modeling scenarios.**

In fact, protein-ligand binding involves a cross-level information transfer process. At the atomic level, intramolecular interactions dictate the chemical structure of the ligand through covalent bonding, which subsequently influences conformational changes at the fragment level. These conformational adjustments ultimately determine the binding mode at the molecular level, affecting the stability of the ligand-protein complex. Meanwhile, intermolecular interactions, such as hydrogen bonding and hydrophobic forces, provide long-range stabilization to the protein-ligand complex. This stabilization can induce conformational adjustments in critical residues within the binding pocket, further reinforcing the stability of intramolecular interactions and ensuring an optimal binding state for the ligand. For example, at the atomic level, covalent interactions determine the stability of chemical bonds within the ligand and influence local electron distribution, thereby adjusting the polarity and flexibility of the ligand. This electronic effect may enhance the likelihood of specific polar functional groups at the substructural level, such as carbonyl (C=O) or amino ($-NH_2$) groups, forming hydrogen bonds or electrostatic interactions, thus facilitating ligand binding to specific residues within the binding pocket. At the molecular level, these intermolecular interactions collectively stabilize the protein-ligand complex and may induce conformational rearrangements in key residues surrounding the binding pocket. Such adjustments optimize the fit of the ligand within the binding site and enhance binding affinity. These hierarchical interactions have not been fully captured by existing models, leading to inherent limitations in feature learning that affect PLBA prediction accuracy. Therefore, accurately capturing the synergistic effects of intramolecular and intermolecular interactions and modeling their interactions at multiple scales is crucial for improving PLBA prediction performance.

To address above challenges, this study introduces an innovative framework—**D**ual-channel **H**ierarchical **I**nteractive **L**earning (DHIL)—which emulates the hierarchical organization of biological systems and explicitly captures both intramolecular and intermolecular interactions, along with their cross-scale coupling, for accurate PLBA predictions. DHIL hierarchically represents the spatial and chemical information of a 3D protein-ligand complex by constructing six chemical entity interaction graphs across two scales: atomic and substructural. These include the ligand atom graph, protein atom graph, atom-atom interaction graph, ligand fragment graph, protein residue graph, and fragment-residue interaction graph, which comprehensively capture protein-ligand interaction information. To fully model both intramolecular and intermolecular interactions, DHIL employs a dual-channel encoding architecture that separately encodes these two types of interactions. The intramolecular channel models covalent interactions within proteins and ligands using atom-level and substructure-level chemical bonding information, while the intermolecular channel learns non-covalent protein-ligand interactions based on spatial distance through heterogeneous atom interaction and substructure interaction graphs. **Beyond independently modeling intramolecular and intermolecular interactions, DHIL introduces a hierarchical interactive learning mechanism to facilitate cross-scale interaction information exchange, thereby capturing hierarchical synergistic effects between these interactions and enhancing affinity prediction performance.** Since the nodes in the two channels are independent and lack explicit associations, we introduce bridge nodes as hubs that enable bidirectional information transfer between distant regular nodes from the two channels. Bridge nodes serve as virtual nodes that connect all regular nodes in the graph, enabling structured information exchange. DHIL incorporates two types of bridge nodes: local bridge nodes, which facilitate atomic-level information exchange within the same substructure across channels, and global bridge nodes, which enable substructure-level information transfer within the same molecule across channels. Through this hierarchical interactive learning mechanism, DHIL progressively optimizes the synergistic representation of intramolecular and intermolecular interactions at different scales, ensuring mutual guidance between the two types of interactions as they integrate from atomic to substructural levels. This approach not only allows for a more precise description of binding mechanisms but also enhances interaction feature representation, ultimately improving the accuracy of PLBA predictions. Finally, DHIL leverages the mutually guided features of intramolecular and intermolecular interactions to construct a richer representation of molecular interactions and further enhance PLBA prediction performance.

To encapsulate the essence of DHIL's innovation, its primary contributions are outlined as follows:

(1) We emphasize the importance of jointly modeling intramolecular and intermolecular interactions, along with their hierarchical synergistic effects across atomic and substructural scales, to provide comprehensive and fine-grained protein-ligand interaction representations for PLBA predictions.

(2) We propose the Dual-channel Hierarchical Interactive Learning (DHIL) framework, which performs dual-channel encoding and hierarchical interactive learning to enable progressive information exchange and complementary guidance between intramolecular and intermolecular interactions, facilitating multi-scale representation learning aligned with the hierarchical nature of biological systems.

(3) We conduct extensive experiments on multiple benchmark datasets, and the results demonstrate that DHIL achieves superior predictive accuracy and generalization ability compared to state-of-the-art baseline methods, validating the effectiveness of capturing hierarchical synergistic effects between the two types of interactions.

The remainder of this paper is organized as follows: Section 2 reviews the recent advances in PLBA predictions, particularly focusing on deep learning-based strategies. Section 3 provides formal definitions and clarifies the graph construction used in the proposed model. Section 4 elaborates on the DHIL framework, detailing the encoding scheme and interaction modeling mechanisms. Section 5 presents extensive experi-

mental evaluations across multiple benchmarks to validate the performance of DHIL. Section 6 discusses the current limitations and outlines potential directions for future research. Finally, Section 7 concludes the paper with a summary of contributions.

2. Related work

In recent years, the rapid development of deep learning offers a novel modeling paradigm for PLBA predictions. In particular, structure-aware models such as GNNs enable efficient representation and interaction modeling of protein and ligand structures in graph space. Current deep learning-based methods for modeling protein-ligand interactions can be broadly categorized into two classes based on the input biological data: separate-data-based and co-structure-based approaches (Li et al., 2023). In separate-data-based approaches (Ballester & Mitchell, 2010; Chu et al., 2022; Gao et al., 2018; Hua et al., 2023; Karimi et al., 2019; Kinnings et al., 2011; Li et al., 2020; Nguyen et al., 2021; Öztürk et al., 2018), proteins and ligands are treated as independent inputs. These models typically preprocess the unbound structural data of proteins and ligands into amino acid sequences and molecular graphs (e.g., SMILES representations (Weininger, 1988)). Convolutional Neural Networks (CNNs), Graph Convolutional Networks (GCNs), and Graph Attention Networks (GATs) are widely employed to encode intrinsic features within covalent bonding structures, thereby extracting both local and global intramolecular interaction information to support downstream affinity prediction tasks. Although these models offer strong feature interpretability and structural clarity, they often fail to capture spatial protein-ligand interaction patterns due to the lack of explicit modeling of the binding conformation. In contrast, co-structure-based methods directly leverage the three-dimensional (3D) structures of protein-ligand complexes to construct graph-based models that incorporate spatial geometry information. With advances in structural biology, especially the advent of the AlphaFold 3 model (Abramson et al., 2024), high-quality 3D complex structures have become increasingly accessible. This progress drives researchers to encode spatial features from the bound conformation into formats such as distance matrices, surface meshes, 3D point clouds, or molecular graphs for deep representation learning (Feinberg et al., 2018; Jiang et al., 2021; Li et al., 2023, 2021; Lim et al., 2019; Stepniewska-Dziubinska et al., 2018; Wang, 2024; Wang et al., 2023; Yan et al., 2024; Zhang et al., 2023, 2022; Zheng et al., 2019). These approaches emphasize the modeling of non-covalent interactions, including hydrogen bonding, hydrophobic forces, and van der Waals interactions, thereby capturing spatial dependencies more faithfully. As a result, co-structure-based modeling has become a central focus in recent PLBA prediction research.

However, whether employing separate-data-based or co-structure-based strategies, most existing models treat intramolecular and intermolecular interactions as independent features, lacking explicit modeling of their hierarchical synergy. This modeling gap may limit the capacity of such models to accurately capture the underlying binding mechanisms. Therefore, integrating the strengths of both interaction types and achieving unified representation and interactive learning across multi-scale graph structures emerge as critical challenges in current PLBA modeling.

3. Definitions

In this section, we delineate the nomenclature employed within DHIL and elucidate the methodology for constructing hierarchical graphs and bridge nodes, in addition to outlining the problem formulation for PLBA estimations. Given the multifaceted nature of graph representation learning addressed in this work, for the sake of clarity, we have compiled a glossary of the primary notations frequently referenced throughout the text in Table 1.

3.1. Hierarchical graph construction

To fully capture the wide range of interactions involved in the protein-ligand binding process, we draw inspiration from the modeling of biomacromolecular structures in (Chu et al., 2022; Gao et al., 2023; Somnath et al., 2021), particularly the hierarchical protein modeling strategies widely adopted in protein-protein interaction (PPI) tasks. In PPI studies, hierarchical modeling integrates information from different levels, such as residues, secondary structures, and functional domains, enhancing the ability to capture complex interaction patterns within proteins. This approach has provided valuable insights for constructing our protein-ligand binding affinity prediction model. By adopting a similar methodology, we incorporate biological entities at different resolutions, enabling our model to represent protein-ligand interactions from multiple perspectives and facilitating a more comprehensive understanding of complex biomolecular mechanisms.

Given a protein-ligand complex in a bound conformation $\mathcal{G}$ as depicted in, which is a stable biological assembly formed by the protein and the ligand component, we define the set of atom nodes as $\mathcal{V} = \mathcal{V}_L^{atom} \cup \mathcal{V}_P^{atom}$ with node position matrix $\mathbf{M} \in \mathbb{R}^{|\mathcal{V}|\times 3}$, where the atom set $\mathcal{V}_L^{atom}$ and $\mathcal{V}_P^{atom}$ represent the composition of two types of atoms of ligand and protein within the complex, respectively. Based on various biochemical information, as shown in Fig. 1(a, b), we decompose a protein-ligand complex $\mathcal{G}$ into six chemical entity interaction graphs across two scales as inputs for the model.

Table 1
Summary of key notations.

Notations	Description
$\mathcal{G}$	3D protein-ligand complex in a bound conformation
$\mathbf{M}$	Node position matrix of $\mathcal{G}$
$\mathcal{V}_L^{atom}$, $\mathcal{V}_P^{atom}$	Atom node sets of ligand and protein components of $\mathcal{G}$, respectively
$\mathcal{G}_L^{atom}$, $\mathcal{E}_L^{atom}$	The ligand atom graph and its edge set
$\mathcal{G}_P^{atom}$, $\mathcal{E}_P^{atom}$	The protein atom graph and its edge set
$\mathcal{G}_I^{atom}$, $\mathcal{V}_I^{atom}$, $\mathcal{E}_I^{atom}$	The atom interaction graph, its node set and edge set
$\mathcal{G}_L^{fra}$, $\mathcal{V}_L^{fra}$, $\mathcal{E}_L^{fra}$	The ligand fragment graph, its node set and edge set
$\mathcal{G}_P^{res}$, $\mathcal{V}_P^{res}$, $\mathcal{E}_P^{res}$	The protein residue graph, its node set and edge set
$\mathcal{G}_I^{sub}$, $\mathcal{V}_I^{sub}$, $\mathcal{E}_I^{sub}$	The substructure interaction graph, its node set and edge set
d_{atom}, d_{res}, d_{sub}	The cutoff threshold for atom interactions, residue interactions and substructure interactions
$\mathcal{V}^{loc_bri}$, $\mathcal{V}_L^{loc_bri}$, $\mathcal{V}_P^{loc_bri}$	The set of local bridge nodes, including those for ligand fragments and protein residues
$\mathcal{V}^{glob_bri}$, $v_L^{glob_bri}$, $v_P^{glob_bri}$	The set of global bridge nodes, global bridge for ligand and global bridge for protein
$\mathcal{T}_{atom}$, $\mathcal{T}_{sub}$	The atom-level information transfer and substructure-level information transfer
$\mathbf{H}_L^{atom}$, $\mathbf{H}_P^{atom}$, $\mathbf{H}_L^{fra}$, $\mathbf{H}_P^{res}$	The node representation matrixes in the intra-type channel for $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$, $\mathcal{G}_L^{fra}$ and $\mathcal{G}_P^{res}$, respectively
$\mathbf{Z}_I^{atom}$, $\mathbf{Z}_I^{sub}$	The node representation matrixes in the inter-type channel for $\mathcal{G}_I^{atom}$ and $\mathcal{G}_I^{sub}$, respectively

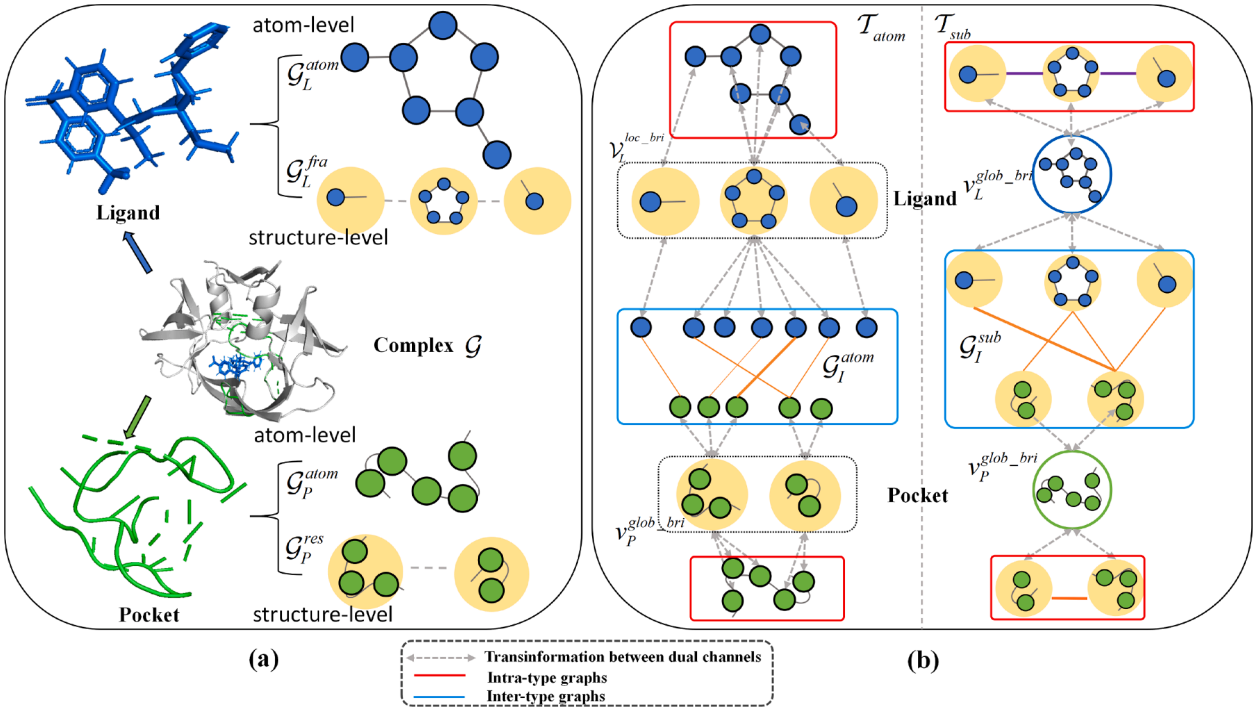


Fig. 1. The illustration of the hierarchical graph construction and interactive learning process in DHIL. (a) The dual-scale hierarchical graph construction for the ligand and the protein pocket, where both components are represented by atom-level and substructure-level graphs. (b) The interactive learning process, which models the information exchange between the two channels at both the atom level (left) and the substructure level (right). During the interaction, local bridge nodes enable bidirectional information exchange at the atom level, while global bridge nodes enable bidirectional information exchange at the substructure level.

3.1.1. Atom-scale graph construction

Ligand Atom Graph. (Intra-type channel) The ligand component, is conceptualized as a molecular graph where atoms serve as nodes and covalent bonds act as edges. We refer to this as the ligand atom graph, defined as $\mathcal{G}_L^{atom} =< \mathcal{V}_L^{atom}, \mathcal{E}_L^{atom} >$, where $\mathcal{E}_L^{atom}$ is the edge set and the edge $e_{L,ij}^{atom}$ notes that the atom $v_{L,i}^{atom}$ and $v_{L,j}^{atom}$ are linked via a covalent bond within $\mathcal{G}$.

Protein Atom Graph.(Intra-type channel) Echoing the establishment process of $\mathcal{G}_L^{atom}$, we delineate the protein atom graph as $\mathcal{G}_P^{atom} =< \mathcal{V}_P^{atom}, \mathcal{E}_P^{atom} >$. Due to the large size of proteins, constructing a complete protein structure can be noisy and time-consuming. Therefore, in our work, we only sample all the atoms contained in the protein residues within the binding pocket domain.

Atom Interaction Graph. (Inter-type channel) Given the absence of provided connection information between the protein and ligand in the dataset, and drawing from insights gained from related works (Jiang et al., 2021; Li et al., 2021), we can infer that the noncovalent interactions between protein and ligand atoms are influenced by their Euclidean distance. In our approach, we make the assumption that the interaction distance between atoms should be below a tunable cutoff threshold, denoted as d_{atom}. Based on this assumption, we construct the atom interaction graph $\mathcal{G}_I^{atom} =< \mathcal{V}_I^{atom}, \mathcal{E}_I^{atom} >$ using the spatial coordinates of the atoms. In this bipartite graph, every edge establishes a connection between an atom of the ligand and an atom of the protein, where the node set is delineated as $\mathcal{V}_I^{atom} = \mathcal{V}_L^{atom} \cup \mathcal{V}_P^{atom}$. The edge set $\mathcal{E}_I^{atom}$ is defined as:

$$\mathcal{E}_I^{atom} = \Big\{ (v_{L,i}^{atom}, v_{P,j}^{atom}) \mid (v_{L,i}^{atom}, v_{P,j}^{atom}) \in \mathcal{V}_L^{atom} \times \mathcal{V}_P^{atom}, \text{ and } Dist(v_{L,i}^{atom}, v_{P,j}^{atom}) \le d_{atom} \Big\}, \tag{1}$$

where $Dist(\cdot,\cdot)$ calculates the spatial Euclidean distance between two atoms, that is, if the euclidean distance between atom $v_{L,i}^{atom}$ in the ligand and atom $v_{P,j}^{atom}$ in the protein is less than the cutoff threshold d_{atom}, an edge from $v_{L,i}^{atom}$ to $v_{P,j}^{atom}$ is added to $\mathcal{E}_I^{atom}$.

3.1.2. Substructure-scale graph construction

Ligand Fragment Graph. (Intra-type channel) The ligand component can also be viewed as being composed of chemically meaningful fragments (e.g., benzene rings, hydroxyl groups, etc.), which serve as the structural basis for protein-ligand interactions. To obtain the fragments of the ligand component, we map the ligand InChI identifiers from the benchmark dataset to SMILES structure data (downloaded from http://ligand-expo.rcsb.org/ld-download.html) and apply BRICS (Degen et al., 2008) for bond-breaking and fragmentation.

We then define the ligand fragment graph as $\mathcal{G}_L^{fra} = \langle \mathcal{V}_L^{fra}, \mathcal{E}_L^{fra} \rangle$, where $\mathcal{V}_L^{fra}$ is the set of nodes, with each node representing a fragment generated by breaking specific single bonds according to the BRICS rules. The edge set $\mathcal{E}_L^{fra}$ reflects the potential recombination relationships between fragments: an edge exists between two fragment nodes $v_{L,i}^{fra}$ and $v_{L,j}^{fra}$ if they can react or be recombined under a valid BRICS reaction type, and $i \neq j$.

Notably, a fragment can also be considered as a subgraph in the ligand atom graph $\mathcal{G}_L^{atom}$, which we denote as $\mathcal{G}_{L,sub_i}^{atom} = (\mathcal{V}_{L,sub_i}^{atom}, \mathcal{E}_{L,sub_i}^{atom})$. Here, $\mathcal{G}_{L,sub_i}^{atom}$ represents the subgraph corresponding to the fragment $v_{L,i}^{fra}$ in $\mathcal{G}_L^{atom}$, and $\mathcal{V}_{L,sub_i}^{atom}$ and $\mathcal{E}_{L,sub_i}^{atom}$ are the sets of nodes and edges, respectively, whose elements constitute the fragment $v_{L,i}^{fra}$.

Protein Residue Graph. (Intra-type channel) Protein residues are the fundamental units of protein structure and function. The sequence of residues (i.e., the protein sequence) is referred to as the primary structure of a protein. However, the primary structure alone is insufficient to understand the hierarchical structure and biological functions of proteins. To capture the interactions between residues formed by the folding of the polypeptide chain, we define the protein residue graph as $\mathcal{G}_P^{res} =< \mathcal{V}_P^{res}, \mathcal{E}_P^{res} >$ where $\mathcal{V}_P^{res}$ is the set of nodes, each representing a residue, extracted from the protein sample using Biopython (Cock et al., 2009). $\mathcal{E}_P^{res}$ is the set of edges, where the presence or absence of an edge between a pair of residues is determined by the spatial distance between their C_a atoms. For this purpose, we also set a cutoff distance d_{res}, that

is:

$$\mathcal{E}_P^{res} = \left\{ (v_{P,i}^{res}, v_{P,j}^{res}) \mid Dist(C_{a,i}, C_{a,j}) \le d_{res} \text{ and } i \neq j \right\}, \tag{2}$$

where $C_{a,i}$ and $C_{a,j}$ represent the C_a atom of residues $v_{P,i}^{res}$ and $v_{P,j}^{res}$, respectively. Similarly, a residue can also be considered as a subgraph in the protein atom graph $\mathcal{G}_P^{atom}$, denoted as $\mathcal{G}_{P,sub_i}^{atom} = (\mathcal{V}_{P,sub_i}^{atom}, \mathcal{E}_{P,sub_i}^{atom})$. Here, $\mathcal{G}_{P,sub_i}^{atom}$ represents the subgraph corresponding to the residue $v_{P,i}^{res}$ in $\mathcal{G}_P^{atom}$, and $\mathcal{V}_{P,sub_i}^{atom}$ and $\mathcal{E}_{P,sub_i}^{atom}$ are the sets of nodes and edges that constitute the residue $v_{P,i}^{res}$. It is worth mentioning that, as previously mentioned when constructing the protein atom graph, we only consider residues within the binding pocket. Specifically, a residue is included in the binding pocket if at least one atom in the residue is within a spatial distance of 10 Å (refer to (Jiang et al., 2021)) from any atom of the ligand.

Substructure Interaction Graph. (Inter-type channel) To capture the interactions between protein residues and ligand fragments, we construct a substructure interaction graph. This graph models the non-covalent interactions between residues and fragments, which are crucial for understanding the protein-ligand binding mechanism. We define the substructure interaction graph as $\mathcal{G}_I^{sub} =< \mathcal{V}_I^{sub}, \mathcal{E}_I^{sub} >$, where the node set $\mathcal{V}_I^{sub}$ includes ligand fragments and protein residues, i.e., $\mathcal{V}_I^{sub} = \mathcal{V}_L^{fra} \cup \mathcal{V}_P^{res}$. The edge set $\mathcal{E}_I^{sub}$ captures the interactions between these substructures based on a cutoff distance d_{sub} , defined as follows:

$$\mathcal{E}_I^{sub} = \left\{ (v_{L,i}^{fra}, v_{P,j}^{res}) \mid \ d(v_{L,i}^{fra}, v_{P,j}^{res}) \le d_{sub} \right\}, \tag{3}$$

where $d(v_{L,i}^{fra}, v_{P,j}^{res})$ denotes the spatial distance between the fragment $v_{L,i}^{fra}$ and the residue $v_{P,j}^{res}$, which can be furtherly defined as:

$$d(v_{L,i}^{fra}, v_{P,j}^{res}) = Min\Big(\{ Dist(v_{L,m}^{atom}, v_{P,n}^{atom}) \mid v_{L,m}^{atom} \in \mathcal{V}_{L,sub_i}^{atom}, \text{ and } v_{P,n}^{atom} \in \mathcal{V}_{P,sub_j}^{atom} \} \Big), \tag{4}$$

where Min denotes the operation of taking the minimum value from a set. It assists in choosing the smallest distance value among all atom pairs between the fragment $v_{L,i}^{fra}$ and the residue $v_{P,j}^{res}$.

3.2. Bridge nodes

In DHIL, to facilitate information exchange between remotely located nodes from intra and inter-type channels, as shown in Fig. 1(b), we define two types of virtual bridge nodes as hubs to enable information communication between individual nodes, namely local bridge nodes, which facilitate atom-level information exchange within the same substructure across the two channels; and global bridge nodes, which assist in substructure-level information exchange within the same molecule across the two channels.

3.2.1. Local bridge nodes

The local bridge nodes can be regarded as the local nodes for substructures of ligands or proteins, assisting in the communication of information between ligand or protein atom nodes in $\mathcal{G}_I^{atom}$ of the inter-type channel and their respective ligand or protein atom nodes in the $\mathcal{G}_L^{atom}$ or $\mathcal{G}_P^{atom}$ of the intra-type channel. We define a set of local bridge nodes for a complex as $\mathcal{V}^{loc_bri} = \mathcal{V}_L^{loc_bri} \cup \mathcal{V}_P^{loc_bri}$, where $\mathcal{V}_L^{loc_bri} = \{v_{L,i}^{loc_bri}\}_{i=1}^{|\mathcal{V}_L^{fra}|}$ and $\mathcal{V}_P^{loc_bri} = \{v_{P,j}^{loc_bri}\}_{j=1}^{|\mathcal{V}_P^{res}|}$ are the local bridge nodes that assist ligand and protein atoms in communicating within the two channels, respectively. Given a local bridge node $v_{L,i}^{loc_bri}$, which corresponds to a ligand fragment $v_{L,i}^{fra}$, it facilitates the communication between atoms within the fragment $\mathcal{V}_{L,sub_i}^{atom}$ in $\mathcal{G}_L^{atom}$ and the corresponding atoms $\mathcal{V}_{I,sub_{L,i}}^{atom}$ in $\mathcal{G}_I^{atom}$. Similarly, for a local bridge node $v_{P,j}^{loc_bri}$, corresponding to a protein residue $v_{P,j}^{res}$, it enables communication between atoms within the residue $\mathcal{V}_{P,sub_j}^{atom}$ in $\mathcal{G}_P^{atom}$ and the corresponding atoms $\mathcal{V}_{I,sub_{P,j}}^{atom}$ in $\mathcal{G}_I^{atom}$.

Formally, the atom-level information transfer $\mathcal{T}_{atom}$ can be defined as:

$$\mathcal{T}_{atom} = \left\{ \begin{matrix} \mathcal{V}_{L,sub_i}^{atom} \leftrightarrow v_{L,i}^{loc_bri} \leftrightarrow \mathcal{V}_{I,sub_{L,i}}^{atom}, \\ \mathcal{V}_{P,sub_j}^{atom} \leftrightarrow v_{P,j}^{loc_bri} \leftrightarrow \mathcal{V}_{I,sub_{P,j}}^{atom} \end{matrix} \right\}, \tag{5}$$

where each local bridge node $v_{L,i}^{loc_bri}$ (or $v_{P,j}^{loc_bri}$) acts as a virtual hub that connects the atom set $\mathcal{V}_{L,sub_i}^{atom}$ (or $\mathcal{V}_{P,sub_j}^{atom}$) from the intra-type channel with its corresponding atom set $\mathcal{V}_{I,sub_{L,i}}^{atom}$ (or $\mathcal{V}_{I,sub_{P,j}}^{atom}$) from the inter-type channel. The symbol $\leftrightarrow$ denotes bidirectional message passing between connected components across channels.

3.2.2. Global bridge nodes

The global bridge nodes can be regarded as global nodes for molecules of ligands or proteins, facilitating communication between fragment nodes or residue nodes in $\mathcal{G}_I^{sub}$ of the inter-type channel and their respective fragment nodes or residue nodes in $\mathcal{G}_L^{fra}$ or $\mathcal{G}_P^{res}$ of the intra-type channel. We define a set of global bridge nodes for a complex as $\mathcal{V}^{glob_bri} = \{v_L^{glob_bri}, v_P^{glob_bri}\}$, where $v_L^{glob_bri}$ and $v_P^{glob_bri}$ are the global bridge nodes that assist all ligand fragments or protein residues in corresponding communicating between the two channels, respectively.

Formally, the substructure-level information transfer $\mathcal{T}_{sub}$ can be defined as:

$$\mathcal{T}_{sub} = \left\{ \begin{matrix} \mathcal{V}_L^{fra} \leftrightarrow v_L^{glob_bri} \leftrightarrow \mathcal{V}_{I,L}^{sub}, \\ \mathcal{V}_P^{res} \leftrightarrow v_P^{glob_bri} \leftrightarrow \mathcal{V}_{I,P}^{sub} \end{matrix} \right\}, \tag{6}$$

where the global bridge node $v_L^{glob_bri}$ (or $v_P^{glob_bri}$) acts as a virtual hub that connects the fragment set $\mathcal{V}_L^{fra}$ (or residue set $\mathcal{V}_P^{res}$) from the intra-type channel with its corresponding substructure set $\mathcal{V}_{I,L}^{sub}$ (or $\mathcal{V}_{I,P}^{sub}$) from the inter-type channel.

3.3. Problem statement

The goal of this study is to predict the binding affinity of a protein-ligand complex using its 3D structure. Let $\mathcal{G}$ denote the 3D complex, and let $\mathbf{M} \in \mathbb{R}^{|\mathcal{V}|\times 3}$ represent the spatial coordinates of all atoms in $\mathcal{G}$.

To comprehensively model multiscale molecular interactions, we construct six graphs from $\mathcal{G}$ and $\mathbf{M}$:

- $\mathcal{G}_L^{atom}$ and $\mathcal{G}_P^{atom}$: atom-level intra-molecular graphs for the ligand and protein;
- $\mathcal{G}_I^{atom}$: atom-level inter-molecular interaction graph;
- $\mathcal{G}_L^{fra}$ and $\mathcal{G}_P^{res}$: substructure-level graphs based on ligand fragments and protein residues;
- $\mathcal{G}_I^{sub}$: substructure-level inter-molecular interaction graph.

With these graphs as input, we aim to learn a predictive function $f(\cdot)$ that estimates the binding affinity $\hat{y}$ of the complex:

$$\hat{y} = f(\mathcal{G}_L^{atom}, \mathcal{G}_P^{atom}, \mathcal{G}_I^{atom}, \mathcal{G}_L^{fra}, \mathcal{G}_P^{res}, \mathcal{G}_I^{sub}). \tag{7}$$

4. Methodology

In this section, we provide a detailed explanation of the DHIL framework. Firstly, in Section 4.1, we offer an overall description of the model in conjunction with Fig. 2 of the DHIL workflow. Next, in Section 4.2, we provide a detailed and formal explanation of the dual-channel encoding. Following this, in Section 4.3, we elaborate on the hierarchical interactive learning, again with a detailed and formal explanation. Finally, in Section 4.4, we introduce the prediction and training methods of DHIL.

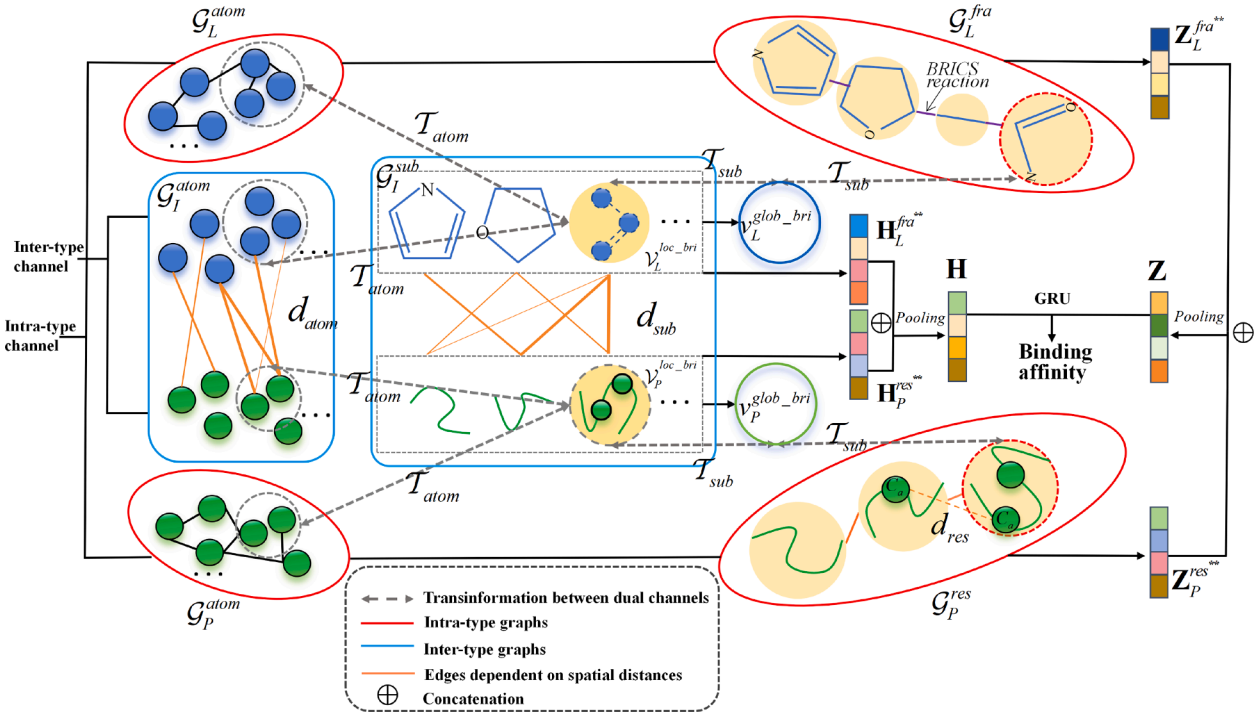


Fig. 2. The workflow of the proposed DHIL.

4.1. Overview

Fig. 2 provides a detailed illustration of the DHIL framework we have proposed. The DHIL model, designed for PLBA prediction, begins by stratifying the 3D protein-ligand complex into six interaction graphs, encompassing both atom and substructure scales. Subsequently, a dual-channel encoding strategy is used to encapsulate the complex interactions: the intra-type channel is dedicated to capturing intramolecular interactions within the individual protein or ligand, whereas the inter-type channel concentrates on the intermolecular interactions occurring between the protein and ligand, taking advantage of the spatial proximities across diverse entities. To amalgamate these two types of interactions, local and global bridge nodes are introduced to facilitate bidirectional information exchange between channels at both atom and substructure levels, respectively. In the final step, DHIL employs the integrated interaction information from both channels to predict PLBAs We summarize the overall procedure of DHIL in Algorithm 1.

4.2. Dual-channel encoding

To comprehensively model the intramolecular and intermolecular interactions of ligand and protein components, a dual-channel encoding network is designed as the backbone. In the intra-type channel, our goal is to perform graph representation learning on $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$ and $\mathcal{G}_L^{fra}$ to obtain latent representations of nodes that perceive the intramolecular covalent chemical environment. Additionally, $\mathcal{G}_P^{res}$ is used to capture the spatial proximity between protein residues based on the distance threshold d_{res}, providing complementary structural context information. In the inter-type channel, the heterogeneous interaction information between protein and ligand components is encoded in $\mathcal{G}_I^{atom}$ and $\mathcal{G}_I^{sub}$ to obtain latent representations of nodes based on spatial distances.

4.2.1. Intra-type channel

Previous works (Li et al., 2023, 2020; Nguyen et al., 2023) have demonstrated the exceptional performance of Graph Attention Network (GAT) (Veličković et al., 2018) in PLBA prediction tasks. Generally, various GAT variants take the topological structure of the graph as well as the physical and chemical attributes of nodes, such as atomic type, partial charge, electrostatic potential, hydrogen bond donor/acceptor status, hydrophobicity, and van der Waals radius, extracted by the open-source cheminformatics software RDKit, as the raw input. These atomic-level physicochemical properties play a crucial role in capturing intermolecular interactions. Atomic type determines the fundamental chemical identity of an atom, influencing its bonding patterns. Partial charge and electrostatic potential reflect charge distribution, which governs electrostatic interactions, a key factor in ligand binding. Hydrogen bond donor/acceptor status is critical for identifying hydrogen bond interactions, which contribute significantly to binding stability. Hydrophobic-

Algorithm 1 The overall procedure for DHIL.

Input: $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$, $\mathcal{G}_I^{atom}$, $\mathcal{G}_L^{fra}$, $\mathcal{G}_P^{res}$, and $\mathcal{G}_I^{sub}$ derived from protein-ligand complex graph $\mathcal{G}$.
Output: Binding affinity $\hat{y}$ for protein-ligand complex graph $\mathcal{G}$.
// Atom-level encoding and interaction
Perform intra-type channel encoding on $\mathcal{G}_L^{atom}$ and $\mathcal{G}_P^{atom}$ for l_{intra} iterations to get $\mathbf{H}_L^{atom^*}$ and $\mathbf{H}_P^{atom^*}$;
Perform inter-type channel encoding on $\mathcal{G}_I^{atom}$ for l_{inter} iterations to get $\mathbf{Z}_I^{atom^*}$;
Perform atom-level interactive learning on $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$, and $\mathcal{G}_I^{atom}$ for l_{atom} iterations to get $\mathbf{H}_L^{atom^{**}}$, $\mathbf{H}_P^{atom^{**}}$, $\mathbf{Z}_L^{atom^{**}}$ and $\mathbf{Z}_P^{atom^{**}}$;
// Substructure-level encoding and interaction
Perform intra-type channel encoding on $\mathcal{G}_L^{fra}$ and $\mathcal{G}_P^{res}$ for l_{intra} iterations to get $\mathbf{H}_L^{fra^*}$ and $\mathbf{H}_P^{res^*}$;
Perform inter-type channel encoding on $\mathcal{G}_I^{sub}$ for l_{inter} iterations to get $\mathbf{Z}_I^{sub^*}$;
Perform substructure-level interactive learning on $\mathcal{G}_L^{fra}$, $\mathcal{G}_P^{res}$, and $\mathcal{G}_I^{sub}$ for l_{sub} iterations to get $\mathbf{H}_L^{fra^{**}}$, $\mathbf{H}_P^{res^{**}}$, $\mathbf{Z}_L^{fra^{**}}$ and $\mathbf{Z}_P^{res^{**}}$;
Perform concatenation and pooling operations on $\mathbf{H}_L^{fra^{**}}$ and $\mathbf{H}_P^{res^{**}}$, and on $\mathbf{Z}_L^{fra^{**}}$ and $\mathbf{Z}_P^{res^{**}}$, respectively, to obtain the final representations $\mathbf{H}$ and $\mathbf{Z}$.
Predict the binding affinity $\hat{y}$ based on $\mathbf{H}$ and $\mathbf{Z}$ by GRU.

ity affects molecular solubility and dictates whether interactions occur in hydrophobic or hydrophilic environments. Van der Waals radius defines steric hindrance and molecular packing, influencing the complementarity between ligand and protein binding sites.

Using the attention mechanism, GAT iteratively assigns different weights to neighboring nodes to aggregate their features, thereby better capturing the relationships and significance between nodes to update the target node representations. This process ultimately outputs a high-level, compact representation of the nodes. In our work, specifically in the intra-type channel network, nodes are allowed to send attention messages to each other along the edges between them, and then each node aggregates all the messages sent to it. The node update process in $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$, $\mathcal{G}_L^{fra}$ and $\mathcal{G}_P^{res}$ can be summarized as follows:

$$\mathbf{H}_i^{(l+1)} = \sigma(\sum_{j\in N_i} a_{ij}\mathbf{W}_1^{(l+1)}\mathbf{H}_j^{(l)} + \mathbf{b}^{(l+1)}), \tag{8}$$

$$a_{ij} = \frac{exp(\sigma(\mathbf{a}_{ij}^{l+1}\cdot\left[\mathbf{W}_2^{(l+1)}\mathbf{H}_i^{(l)} \parallel \mathbf{W}_3^{(l+1)}\mathbf{H}_j^{(l)}\right]))}{\sum_{k\in N_i} exp(\sigma(\mathbf{a}_{ik}^{l}\cdot\left[\mathbf{W}_2^{(l+1)}\mathbf{H}_i^{(l)} \parallel \mathbf{W}_3^{(l+1)}\mathbf{H}_k^{(l)}\right]))}, \tag{9}$$

where σ is the $LeakyReLU$ function, $\mathbf{W}^{(l+1)} \in \mathbb{R}^{d\times d}$, $\mathbf{a}^{(l+1)} \in \mathbb{R}^d$ and $\mathbf{b}^{(l+1)} \in \mathbb{R}^d$ are $(l+1)$-th layer's trainable weights and bias in the intra-type channel layer, respectively, and d is the dimensionality of embedding features. a_{ij} is the attentional coefficient normalized by Softmax function computed as Eq. (9) to account for the relevance of node i and j, $\cdot$ is the inner product and $\parallel$ is the concatenation operation. $\mathbf{H}_i^{(l)} = \mathbf{H}^{(l)}[i,:] \in \mathbb{R}^d$ represents the i-th representation of node in feature matrix $\mathbf{H}^{(l)}$ in the l-th layer.

For the initial features of substructure nodes $\mathbf{H}_L^{fra^{(0)}} \in \mathbb{R}^{\left|\mathcal{V}_L^{fra}\right|\times d}$ and $\mathbf{H}_P^{res^{(0)}} \in \mathbb{R}^{\left|\mathcal{V}_P^{res}\right|\times d}$ in $\mathcal{G}_L^{fra}$ and $\mathcal{G}_P^{res}$, we first extract its geometric and chemical characteristics, including its dihedral angles, volume, polarity, charge, hydropathy, secondary structure, solvent accessibility, and hydrogen bond interactions, to obtain the physicochemical embedding $\mathbf{x}_i \in \mathbb{R}^d$. These substructure-level physicochemical properties are essential for capturing macroscopic interaction patterns that extend beyond atomic interactions. Dihedral angles determine the conformational flexibility of ligand fragments and protein residues, influencing steric compatibility within binding sites, while volume constraints dictate spatial complementarity and molecular fit. Polarity and charge affect electrostatic interactions and hydrogen bonding, both of which are fundamental to protein-ligand affinity. Hydropathy influences whether substructures prefer hydrophobic or hydrophilic environments, playing a key role in ligand binding selectivity. Secondary structure provides a structural framework for residues, affecting how binding pockets accommodate ligands, while solvent accessibility determines whether a residue or ligand fragment is buried within the protein or exposed to the solvent, impacting its likelihood of interaction. Hydrogen bond interactions contribute significantly to the stability of protein-ligand complexes, dictating specificity and affinity. Then, the substructure feature $\mathbf{x}_i$ is concatenated with the sum of final atom-level embeddings after atom-level interactive learning (see Section 3.3.1 for details) within the substructure to serve as the initial representation of the substructure node:

$$\mathbf{H}_{L,i}^{fra^{(0)}} = Concat(\mathbf{x}_{L,i}, \sum_{k\in\mathcal{V}_{L,sub_i}^{atom}} \mathbf{H}_{L,k}^{atom^{**}}), \tag{10}$$

$$\mathbf{H}_{P,j}^{res^{(0)}} = Concat(\mathbf{x}_{L,j}, \sum_{k\in\mathcal{V}_{P,sub_j}^{atom}} \mathbf{H}_{P,k}^{atom^{**}}), \tag{11}$$

where $\mathbf{H}_{L,i}^{fra^{(0)}}$ and $\mathbf{H}_{P,j}^{res^{(0)}}$ are the initial representations of fragment $v_{L,i}^{fra}$ in $\mathcal{G}_L^{fra}$ and residue $v_{P,j}^{res}$ in $\mathcal{G}_P^{res}$, respectively, while $\mathbf{x}_{L,i} \in \mathbb{R}^d$ and $\mathbf{x}_{L,j} \in \mathbb{R}^d$ are the extracted physicochemical representations of these two nodes. $\mathbf{H}_{L,k}^{atom^{**}}$ and $\mathbf{H}_{P,k}^{atom^{**}}$ are the final enhanced representations after atom-level interactive learning for atoms in $\mathcal{G}_L^{atom}$ and $\mathcal{G}_P^{atom}$, respectively.

In summary, after l_{intra} iterations, the outputs of our intra-type channel correspond to the node representations $\mathbf{H}_L^{atom^*} \in \mathbb{R}^{\left|\mathcal{V}_L^{atom}\right|\times d}$, $\mathbf{H}_P^{atom^*} \in \mathbb{R}^{\left|\mathcal{V}_P^{atom}\right|\times d}$, $\mathbf{H}_L^{fra^*} \in \mathbb{R}^{\left|\mathcal{V}_L^{fra}\right|\times d}$ and $\mathbf{H}_P^{res^*} \in \mathbb{R}^{\left|\mathcal{V}_P^{res}\right|\times d}$ of the graphs $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$,$\mathcal{G}_L^{fra}$ and $\mathcal{G}_P^{res}$, respectively.

4.2.2. Inter-type channel

In the inter-type channel, we aim to extract noncovalent interactions between protein and ligand components represented by $\mathcal{G}_I^{atom}$ and $\mathcal{G}_I^{sub}$. Since protein-ligand binding occurs in three-dimensional space, incorporating spatial distance between heterogeneous entities is essential for accurately modeling intermolecular interactions. To this end, we construct two distance-dependent interaction matrices: $\mathbf{D}^{atom} \in \mathbb{R}^{|\mathcal{V}_L^{atom}|\times|\mathcal{V}_P^{atom}|}$ and $\mathbf{D}^{sub} \in \mathbb{R}^{|\mathcal{V}_L^{sub}|\times|\mathcal{V}_P^{sub}|}$, defined as follows:

$$\mathbf{D}_{ij}^{atom} = \begin{cases} \frac{\exp(d_{atom}-\mathrm{Dist}(v_{L,i}^{atom},v_{P,j}^{atom}))}{\sum_{v_{P,k}^{atom}\in\mathcal{N}(v_{L,i}^{atom})}\exp(d_{atom}-\mathrm{Dist}(v_{L,i}^{atom},v_{P,k}^{atom}))}, & \text{if } \mathrm{Dist}(v_{L,i}^{atom},v_{P,j}^{atom}) \le d_{atom} \\ 0, & \text{otherwise} \end{cases}, \tag{12}$$

$$\mathbf{D}_{ij}^{sub} = \begin{cases} \frac{\exp(d_{sub}-d(v_{L,i}^{fra},v_{P,j}^{res}))}{\sum_{v_{P,k}^{res}\in\mathcal{N}(v_{L,i}^{fra})}\exp(d_{sub}-d(v_{L,i}^{fra},v_{P,k}^{res}))}, & \text{if } d(v_{L,i}^{fra},v_{P,j}^{res}) \le d_{sub} \\ 0, & \text{otherwise} \end{cases}, \tag{13}$$

where $\mathbf{D}_{ij}^{atom}$ and $\mathbf{D}_{ij}^{sub}$ represent the normalized distance-based interaction weights (computed via a Softmax function) between ligand atom $v_{L,i}^{atom}$ and protein atom $v_{P,j}^{atom}$ in $\mathcal{G}_I^{atom}$, and between ligand fragment $v_{L,i}^{fra}$ and protein residue $v_{P,j}^{res}$ in $\mathcal{G}_I^{sub}$, respectively. $\mathcal{N}(v_{L,i}^{atom})$ and $\mathcal{N}(v_{L,i}^{fra})$ denote the sets of neighboring protein atoms or residues within their respective cutoff distances.

Based on these interaction matrices, we update the node representations in the inter-type channel by aggregating information from their cross-molecular counterparts. Specifically, each ligand node receives interaction-weighted messages from all relevant protein nodes of the same scale, and vice versa. The updated representations are given by:

$$\mathbf{Z}_{L,i}^{atom^{(l+1)}} = \sum_{j=1}^{|\mathcal{V}_P^{atom}|} \sigma\left(\mathbf{D}_{ij}^{atom}\cdot\mathbf{W}_4^{(l+1)}\left(\mathbf{Z}_{L,i}^{atom^{(l)}}\odot\mathbf{Z}_{P,j}^{atom^{(l)}}\right)\right), \tag{14}$$

$$\mathbf{Z}_{L,i}^{fra^{(l+1)}} = \sum_{j=1}^{|\mathcal{V}_P^{res}|} \sigma\left(\mathbf{D}_{ij}^{sub}\cdot\mathbf{W}_5^{(l+1)}\left(\mathbf{Z}_{L,i}^{fra^{(l)}}\odot\mathbf{Z}_{P,j}^{res^{(l)}}\right)\right), \tag{15}$$

where $\mathbf{W}_4^{(l+1)}, \mathbf{W}_5^{(l+1)} \in \mathbb{R}^{d\times d}$ are learnable projection matrices, $\odot$ denotes element-wise multiplication, and σ is a nonlinear activation function. Here, $\mathbf{Z}_{L,i}^{fra^{(l)}}$ and $\mathbf{Z}_{P,j}^{res^{(l)}}$ refer to the substructure-level node representations in $\mathcal{G}_I^{sub}$. Note that the initial features of nodes in $\mathcal{G}_I^{atom}$ and $\mathcal{G}_I^{sub}$ are inherited from the intra-type channel outputs $\mathbf{H}_L^{atom^*}$, $\mathbf{H}_P^{atom^*}$, $\mathbf{H}_L^{fra^*}$, and $\mathbf{H}_P^{res^*}$, which are the encoded features of $\mathcal{G}_L^{atom}$, $\mathcal{G}_P^{atom}$, $\mathcal{G}_L^{fra}$, and $\mathcal{G}_P^{res}$, respectively.

In summary, after l_{inter} iterations, the outputs of inter-type channel are $\mathbf{Z}_I^{atom^*} \in \mathbb{R}^{\left|\mathcal{V}_I^{atom}\right|\times d}$ and $\mathbf{Z}_I^{sub^*} \in \mathbb{R}^{\left|\mathcal{V}_I^{sub}\right|\times d}$ for $\mathcal{G}_I^{atom^*}$ and $\mathcal{G}_I^{sub}$, respectively, where $\mathbf{Z}_I^{atom^*} = \mathbf{Z}_L^{atom^*} \parallel \mathbf{Z}_P^{atom^*}$ and $\mathbf{Z}_I^{sub^*} = \mathbf{Z}_L^{fra^*} \parallel \mathbf{Z}_P^{res^*}$.

4.3. Hierarchical interactive learning

To encapsulate the intricate interplay between the two forms of interactions, we have developed a hierarchical interactive learning scheme that enhances information exchange between the two channels at both atom and substructure-levels. Specifically, we introduce bridge nodes

as hubs to connect independent regular nodes in both channels, enabling remote bidirectional information communication between distant nodes. We summarize the overall procedure of hierarchical interactive learning in Algorithm 2.

Algorithm 2 The overall procedure for hierarchical interactive learning.

Input: $\mathbf{H}_L^{atom^*}$, $\mathbf{H}_L^{fra^*}$, $\mathbf{H}_P^{atom^*}$ and $\mathbf{H}_P^{res^*}$ from intra-type channel, $\mathbf{Z}_L^{atom^*}$, $\mathbf{Z}_L^{fra^*}$, $\mathbf{Z}_P^{atom^*}$ and $\mathbf{Z}_P^{res^*}$ from inter-type channel.
Output: Enhanced substructure representations $\mathbf{H}_L^{fra^{**}}$, $\mathbf{H}_P^{res^{**}}$, $\mathbf{Z}_L^{fra^{**}}$, $\mathbf{Z}_P^{res^{**}}$.
// Atom-level Interactive Learning (intra-type → inter-type)
for each ligand fragment $v_{L,i}^{fra}$ and protein residue $v_{P,i}^{res}$ **do**
  Initialize local bridge node features;
  for $l = 0$ to $l_{atom} - 1$ **do**
    Gather information from intra-type atoms to local bridge node;
    Update local bridge node using warp gate mechanism;
    Apply GRU to update local bridge node features;
    Transfer information from local bridge node to inter-type atom nodes;
    Update inter-type atom nodes using warp gate and GRU;
  end for
end for
Obtain enhanced atom representations: $\mathbf{Z}_L^{atom^{**}}$, $\mathbf{Z}_P^{atom^{**}}$.
// Atom-level Interactive Learning (inter-type → intra-type)
Obtain enhanced atom representations: $\mathbf{H}_L^{atom^{**}}$, $\mathbf{H}_P^{atom^{**}}$.
// Substructure-level Interactive Learning (intra-type → inter-type)
Initialize global bridge node features;
for $l = 0$ to $l_{sub} - 1$ **do**
  Gather information from intra-type substructures to global bridge node;
  Update global bridge node using warp gate mechanism;
  Apply GRU to update global bridge node features;
  Transfer information from global bridge node to inter-type substructure nodes;
  Update inter-type substructure nodes using warp gate and GRU;
end for
Obtain enhanced substructure representations: $\mathbf{Z}_L^{fra^{**}}$, $\mathbf{Z}_P^{res^{**}}$.
// Substructure-level Interactive Learning (inter-type → intra-type)
Obtain enhanced substructure representations: $\mathbf{H}_L^{fra^{**}}$, $\mathbf{H}_P^{res^{**}}$.
return $\mathbf{H}_L^{fra^{**}}$, $\mathbf{H}_P^{res^{**}}$, $\mathbf{Z}_L^{fra^{**}}$, $\mathbf{Z}_P^{res^{**}}$.

4.3.1. Atom-level interactive learning

The purpose of atom-level interactive learning is to separately facilitate the exchange of information between ligand and protein atom nodes in $\mathcal{G}_I^{atom}$ and their corresponding ligand atom nodes in $\mathcal{G}_L^{atom}$ and protein atom nodes in $\mathcal{G}_P^{atom}$ within the same substructure. Here, to avoid redundant expressions, we detail the unidirectional information transfer (from intra to inter-type channel) between ligand atoms across the two channels, as the bidirectional process follows a similar workflow.

For the local bridge nodes $v_{L,i}^{loc_bri} \in \mathcal{V}_L^{loc_bri}$, before information transfer, its feature $\mathbf{B}_{L,i}^{loc^{(0)}} \in \mathbb{R}^d$ is initialized as the summation of the features of the atoms from $\mathcal{G}_I^{atom}$ encompassed by the corresponding fragment $v_{L,i}^{fra}$, that is,

$$\mathbf{B}_{L,i}^{loc^{(0)}} = \sum_{j=1}^{\left|\mathcal{V}_{I,sub_{L,i}}^{atom}\right|} \mathbf{Z}_{L,j}^{atom^*}. \tag{16}$$

Then, the information transfer process is accomplished through the two steps: transmitting information from regular atom nodes in the intra-type channel to bridge node and transmitting information from bridge node to regular atom nodes in the inter-type channel:

Transmitting information from regular atom nodes in the intra-type channel to local bridge node: We firstly gather information from the regular atom nodes in $\mathcal{G}_L^{atom}$ of the intra-type channel to be sent to the bridge nodes. The information gathered from each regular atom is weighted using the cosine similarity with the bridge node, that is,

$$\mathbf{u}_{a\to b}^{(l+1)} = \sum_{j=1}^{\left|\mathcal{V}_{L,sub_i}^{atom}\right|} \sigma(c_{ij}^{l+1}\mathbf{W}_6^{l+1}\mathbf{H}_{L,j}^{atom^{(*+l)}}), \tag{17}$$

$$c_{ij}^{l+1} = \frac{\mathbf{H}_{L,j}^{atom^{(*+l)}} \cdot \mathbf{B}_{L,i}^{loc^{(l)}}}{\left\|\mathbf{H}_{L,j}^{atom^{(*+l)}}\right\|_2 \left\|\mathbf{B}_{L,i}^{loc^{(l)}}\right\|_2}, \tag{18}$$

where $\mathbf{u}_{a\to b}^{(l+1)} \in \mathbb{R}^d$ is the information sent from all regular atom nodes in the l-th layer to the $(l+1)$-th layer's bridge node $v_{L,i}^{loc_bri}$ for updating in atom-level interactive learning, $\mathbf{W}_8^{l+1} \in \mathbb{R}^{d\times d}$ is the $(l+1)$-th layer's weight matrix, $\mathbf{H}_{L,j}^{atom^{(*+l)}}$ refers to the final output representation of the j-th regular atom node in the intra-type channel, which then undergoes an additional l iterations in the atom-level interactive learning, c_{ij}^{l+1} represents the information strength sent from the j-th regular atom node to the bridge node $v_{L,i}^{loc_bri}$ and $\|\cdot\|_2$ is to compute the two norm value of a vector. Then, to update the representation of the bridge node, inspired by (Ishiguro et al., 2019), we adopt the warp gate mechanism to adaptively control the mixing rate of the information from the atoms in the intra-type channel that is passed to the bridge node and the information from the bridge node itself, as follows:

$$\mathbf{t}_{L,i}^{loc^{(l+1)}} = WG(\mathbf{B}_{L,i}^{loc^{(l)}}, \mathbf{u}_{a\to b}^{(l+1)}), \tag{19}$$

$$WG(\mathbf{B}_{L,i}^{loc^{(l)}}, \mathbf{u}_{a\to b}^{(l+1)}) = (\mathbf{1} - \mathbf{g}_{L,i}^{loc^{(l+1)}}) \odot \mathbf{u}_{a\to b}^{(l+1)} + \mathbf{g}_{L,i}^{loc^{(l+1)}} \odot \mathbf{B}_{L,i}^{loc^{(l)}}, \tag{20}$$

$$\mathbf{g}_{L,i}^{loc^{(l+1)}} = \sigma(\mathbf{W}_7^{(l+1)}\mathbf{B}_{L,i}^{loc^{(l)}} + \mathbf{W}_8^{(l+1)}\mathbf{u}_{a\to b}^{(l+1)}), \tag{21}$$

where WG is the warp gate function, which enables the controlled mixing of branched information flows to obtain the message $\mathbf{t}_{L,i}^{loc^{(l+1)}} \in \mathbb{R}^d$ that is passed to the bridge node, $\mathbf{g}_{L,i}^{loc^{(l+1)}} \in \mathbb{R}^d$ is the tensor of warp gate coefficients for the mixing transmission and $\mathbf{W}_8^{(l+1)}$, $\mathbf{W}_9^{(l+1)} \in \mathbb{R}^{d\times d}$ are weight matrixes.

Based on the received messages passed to the bridge nodes$\mathbf{t}_{L,i}^{loc^{(l+1)}}$, we apply Gated Recurrent Units (GRU) (Cho et al., 2014) to update features of the bridge node, that is,

$$\mathbf{B}_{L,i}^{loc^{(l+1)}} = GRU(\mathbf{B}_{L,i}^{loc^{(l)}}, \mathbf{t}_{L,i}^{loc^{(l+1)}}). \tag{22}$$

Transmitting information from local bridge node to regular atom nodes in the inter-type channel:

After obtaining the updated bridge node representations, to pass information to the regular nodes in $\mathcal{G}_I^{atom}$ in the inter-type channel, we firstly gather information from the bridge nodes through a single-layer neural network, as follows:

$$\mathbf{u}_{b\to a}^{(l+1)} = \sigma(\mathbf{W}_9^{(l+1)}\mathbf{B}_{L,i}^{loc^{(l+1)}}). \tag{23}$$

Then we also adopt the warp gate mechanism to adaptively mix information from the bridge node that is passed to the atoms in the inter-type channel and the information from the atom node itself, that is,

$$\mathbf{t}_{L,i}^{atom^{(*+l+1)}} = WG(\mathbf{Z}_{L,i}^{atom^{(*+l)}}, \mathbf{u}_{b\to a}^{(l+1)}), \tag{24}$$

where $\mathbf{t}_{L,i}^{atom^{(*+l+1)}}$ is the message passed to $v_{L,i}^{atom}$ in $\mathcal{G}_I^{atom}$ at the $(l+1)$-th layer, $\mathbf{Z}_{L,i}^{atom^{(*+l)}}$ refers to the features of the i-th regular atom node in $\mathcal{G}_I^{atom}$ at the l-th layer for atom-level interactive learning. For each regular ligand node in $\mathcal{G}_I^{atom}$, we update its features as follows,

$$\mathbf{Z}_{L,i}^{atom^{(*+l+1)}} = GRU(\mathbf{Z}_{L,i}^{atom^{(*+l)}}, \mathbf{t}_{L,i}^{atom^{(*+l+1)}}). \tag{25}$$

Note that the information transfer process from the inter to the intra-type channel is a completely similar reverse operation, and the atom-level interactive learning process for protein atoms between $\mathcal{G}_P^{atom}$ and $\mathcal{G}_I^{atom}$ follows exactly the same procedure as for the ligand.

In summary, after l_{atom} atom-level interactive learning, we obtain atom representations enhanced by atom-level interactions between the intra-type and inter-type channels, specifically $\mathbf{H}_L^{atom^{**}}$ in $\mathcal{G}_L^{atom}$, $\mathbf{H}_P^{atom^{**}}$ in $\mathcal{G}_P^{atom}$, and $\mathbf{Z}_I^{atom^{**}}$ in $\mathcal{G}_I^{atom}$, where $\mathbf{Z}_I^{atom^{**}} = \mathbf{Z}_L^{atom^{**}} \parallel \mathbf{Z}_P^{atom^{**}}$.

4.3.2. Substructure-level interactive learning

Substructure-level interactive learning aims to facilitate information exchange between ligand and protein nodes in $\mathcal{G}_I^{sub}$ and their corresponding ligand substructure nodes in $\mathcal{G}_L^{fra}$, and protein substructure nodes in $\mathcal{G}_P^{res}$, through global bridge nodes. Global bridge nodes can be seen as global representations of all substructures of ligands or proteins. Substructure-level interactive learning employs a similar information exchange process to atom-level interactive learning, but involves substructure nodes rather than atom nodes. The information transmission process is also two-step (from intra to inter-type channel): transmitting information from regular substructure nodes in the intra-type channel to the global bridge node, and transmitting information from the global bridge node to regular substructure nodes in the inter-type channel.

After l_{sub} iterations of substructure-level interactive learning, we achieve substructure representations enhanced by substructure-level interactions between the intra-type and inter-type channels, specifically $\mathbf{H}_L^{fra^{**}}$ in $\mathcal{G}_L^{fra}$, $\mathbf{H}_P^{res^{**}}$ in $\mathcal{G}_P^{res}$, and $\mathbf{Z}_I^{sub^{**}}$ in $\mathcal{G}_I^{sub}$, where $\mathbf{Z}_I^{sub^{**}} = \mathbf{Z}_L^{fra^{**}} \parallel \mathbf{Z}_P^{res^{**}}$.

4.4. Prediction and training

In this subsection, we pool the atom and substructure-level node representations to obtain the representations of the two channels, respectively. We then use a GRU to fuse the representations of the two channels to predict protein-ligand binding affinity for a complex:

$$\mathbf{H} = Pooling(\mathbf{H}_L^{fra^{**}} \parallel \mathbf{H}_P^{res^{**}}), \tag{26}$$

$$\mathbf{Z} = Pooling(\mathbf{Z}_I^{sub^{**}}), \tag{27}$$

$$\hat{y} = \mathbf{W}_{10}(GRU(\mathbf{H}, \mathbf{Z})), \tag{28}$$

where $\mathbf{H}$ and $\mathbf{Z}$ are the final representations of the intra and inter-type channel, respectively. $Pooling$ is an attention pooling function (Li et al., 2016). Subsequently, following (Öztürk et al., 2018), we determine the loss by evaluating the Mean-Square Error (MSE) between the predicted affinity $\hat{y}$ and the actual value y for the purpose of model optimization:

$$Loss = \frac{1}{n}\sum_{\mathcal{G}\in\mathcal{D}}(\hat{y} - y)^2, \tag{29}$$

where $\mathcal{D}$ encompasses all the training protein-ligand complexes along with their corresponding binding affinities, and n represents the total count of complexes within $\mathcal{D}$.

5. Experiments

This section conducts a comprehensive experimental analysis of the proposed DHIL. First, Section 4.1 presents the experimental settings, including the datasets, baselines, and model parameter configurations. Next, Section 4.2 performs extensive experiments to compare DHIL with baseline methods in terms of overall performance, generalization, and robustness across various data scenarios. Then, Section 4.3 conducts an in-depth data analysis and visualization to demonstrate the effectiveness of DHIL's innovative components. Subsequently, Section 4.4 explores whether DHIL exhibits biochemical interpretability when inferring PLBAs. Finally, Section 4.5 extends the validation to additional datasets to assess the applicability and stability of DHIL across a broader range of data distributions.

5.1. Experiment settings

5.1.1. Datasets

The assessment of all approaches is carried out on two renowned public datasets that are commonly used for PLBA prediction tasks:

PDBbind (Wang et al., 2005) stands out as a well-regarded public dataset, providing intricate 3D structural insights into protein-ligand complexes, complete with experimentally ascertained binding affinities. In the realm of biological experimentation, binding affinity is frequently denoted by dissociation constants (K_d), inhibition constants (K_i) or half-maximal inhibitory concentrations (IC_{50}) (He et al., 2017). The negative logarithms of these metrics are applied in both the PDBbind and the subsequent CSAR-HiQ datasets. For the experimental section of this work, we chiefly employ the PDBbind v2016 dataset, which has been widely used in prior research efforts (Stepniewska-Dziubinska et al., 2018; Zheng et al., 2019). This dataset is subdivided into three categories: the *general* set, the *refined* set, and the *core* set. The *general* set includes a total of 16,179 protein-ligand complexes, whereas the *refined* set is comprised of 4057 complexes, cherry-picked from the *general* set for their enhanced quality. The *core* set, made up of 290 complexes, serves as a high-caliber benchmark that has undergone a rigorous selection procedure (Su et al., 2018b). For the sake of clarity in this paper, we will be referring to the 3767 complexes that distinguish the *refined* from the *core* subsets as the *refined set* of PDBbind.

CSAR-HiQ (Dunbar Jr et al., 2011) serves as an extra benchmarking dataset, which includes two subsets sourced independently: the fist subset with 176 protein-ligand samples and the another one with 167. We employ this additional dataset to assess the model's ability to generalize.

To understand the underlying distribution trends of key data within the datasets and to explore their commonalities and distinctions, we perform a statistical examination of the crucial information, as shown in Fig. 3. First of all, the distribution of ligand atom counts in the complexes reveals a clustering around smaller molecules, typically containing up to 50 atoms, while the distribution of protein atom counts exhibits a wider and more diverse spectrum, as shown in Fig. 3(a) and (b), respectively. Next, the ground-truths for binding affinity in both datasets demonstrate comparable normal distribution patterns (Fig. 3(c)), with P-values of 0.024 and 0.013 as determined by the Kolmogorov-Smirnov (K-S) test for the PDBbind and CSAR-HiQ sets, respectively. Furthermore, we examine the distributions of atomic distances across different molecular interactions, observing that intramolecular interactions are largely focused within a range of 1 Å to 2 Å, while intermolecular interactions reach out to greater distances (Fig. 3(d, e)). This variation presents a challenge in determining a suitable cutoff distance that effectively encapsulates the diverse array of protein-ligand interaction information. Finally, through the calculation of the maximum sequence similarity (Li et al., 2020) between the target proteins in the test set complexes and those in the training set, we observe a notably lower degree of overlap for the CSAR-HiQ set in contrast to the *core set* (Fig. 3(f)). Consequently, we utilize the diverse CSAR-HiQ dataset to test the model's capability for generalization.

To uphold equity, we adhere to the standard data partitioning approach prevalent in prior studies (Ballester & Mitchell, 2010; Li et al., 2015a, 2021; Stepniewska-Dziubinska et al., 2018; Zheng et al., 2019). Specifically, we opt for the *refined set* of PDBbind as the training dataset, as suggested by Ballester and Mitchell (2010). Subsequently, we randomly divide the *refined set* at a 9:1 ratio to form a training and validation subset. We utilize the *core set* and CSAR-HiQ set for testing, meticulously excluding any complexes that have already been part of the *refined set*, which ensures that the testing subsets comprise exclusive samples

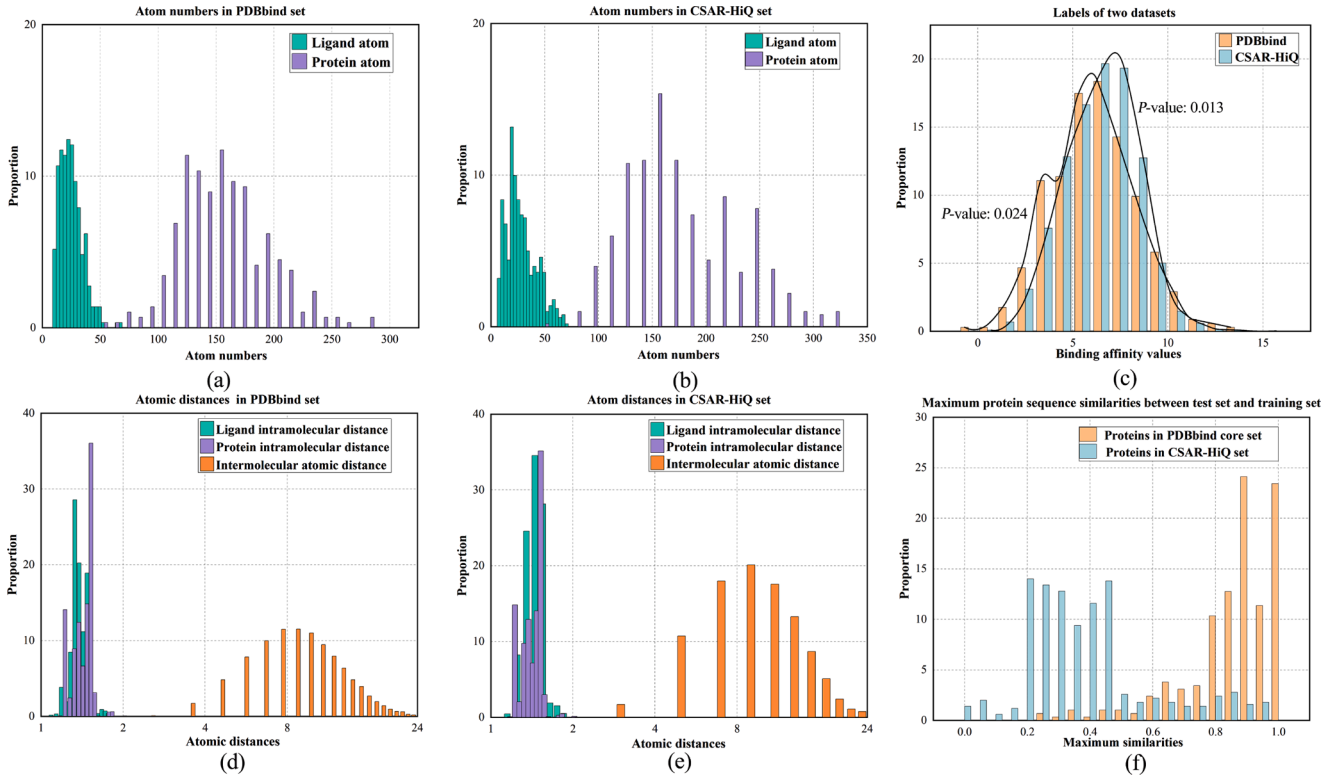


Fig. 3. Histograms illustrating the distribution of critical attributes for the PDBbind and CSAR-HiQ datasets are presented. (a), (b): The frequency distributions of atom counts in ligands and proteins for the two datasets. (c): The frequency distribution of binding affinity label values for both datasets is depicted. To ascertain the statistical significance of the deviations from a normal distribution, a Kolmogorov-Smirnov (K-S) test is performed. The *P*-values for the PDBbind and CSAR-HiQ datasets are found to be 0.024 and 0.013, respectively. (d), (e): The lengths of intramolecular covalent bonds in ligands and proteins, along with intermolecular protein-ligand atomic distances, are depicted across the two datasets. (f): The distribution of the highest sequence similarities between proteins in the test sets (*core set* and CSAR-HiQ set) and the training set (*refined set*) is illustrated.

that are not involved in the training or validation process. Moreover, acknowledging the underlying influence of lower-quality data from the *general set* on model performance (Li et al., 2015a), in training, we conduct an extra experiment employing the lower-quality samples from *general set* to evaluate the model's robustness.

5.1.2. Evaluation metrics

To comprehensively assess DHIL's performance, we employ the regression analysis metrics usually used in Stepniewska-Dziubinska et al. (2018), Zheng et al. (2019), such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Pearson's correlation coefficient (R) and Standard Deviation (SD). These metrics enable us to assess the model's accuracy, precision, and correlation with the actual binding affinity values in a holistic manner.

5.1.3. Benchmarks

To assess the efficacy of the introduced DHIL model, we have chosen the following approaches as comparative benchmarks:

• **ML-based Methods** include linear regression (LR), support vector regression (SVR), and random forest (RF). These methods take the intermolecular interaction features introduced in RF-Score (Ballester & Mitchell, 2010) as input and predict the protein-ligand binding affinity.

• **CNN-based Methods:**

Pafnucy (Stepniewska-Dziubinska et al., 2018) develops a novel deep neural network that estimates ligand-receptor binding affinity by representing the complex with a 3D grid and applying 3D convolution to generate a unified feature map for both protein and ligand atoms.

OnionNet (Zheng et al., 2019) develops a deep convolutional neural network model that uses rotation-free element-pair-specific contacts between ligands and protein atoms, organized into distance ranges to capture both local and nonlocal interaction information.

• **GraphDTA Methods:**

GraphDTA (Nguyen et al., 2021) integrates GNN architectures to decipher complex graph topologies and incorporates CNN algorithms for the analysis of protein sequences. It encompasses four distinct versions, each employing a type of GNN technique: GCN (Kipf & Welling, 2017), GAT (Veličković et al., 2018), GIN (Xu et al., 2019), and a hybrid GAT-GCN model as a blend of GAT and GCN.

• **GNN-based Methods:**

CMPNN (Song et al., 2020) enhances the message interaction between nodes and edges through a communicative kernel and introduces a new message enhancement module to enrich the message generation process, aiming to optimize the embedding of protein-ligand complexes.

IGN (Jiang et al., 2021) stacks two independent graph convolution modules to sequentially learn the intramolecular and intermolecular interactions, and employs the learned intermolecular interactions for the prediction of PLBAs.

SIGN (Li et al., 2021) iteratively executes a node-edge aggregation process to update the embeddings of nodes and edges for modeling protein-ligand complexes, and reflects global interactions by modeling the interaction matrix of heterogeneous edges.

GLI (Zhang et al., 2022) takes into account the high-energy short-range interactions between atoms as well as the low-energy long-range interactions between non-bonded atoms to achieve joint global-local interaction modeling and enhance the performance for the prediction of PLBAs.

SS-GNN (Zhang et al., 2023) develops an edge-based atom-pair feature aggregation method to represent complex interactions and a graph pooling-based method to predict the binding affinity of the complex.

Table 2
GNN-based baseline models for PLBA prediction sorted by year.

Method	Year	Model Type	Reference
CMPNN	2020	GNN	(Song et al., 2020)
GraphDTA	2020	GNN	(Nguyen et al., 2021)
IGN	2021	GNN	(Jiang et al., 2021)
ECIF6: LD-GBT	2021	GNN	(Sánchez-Cruz et al., 2021)
PSH-ML	2021	GNN	(Liu et al., 2021)
SIGN	2021	GNN	(Li et al., 2021)
GLI	2022	GNN	(Zhang et al., 2022)
SS-GNN	2023	GNN	(Zhang et al., 2023)
SybylGGL-Score	2023	GNN	(Rana & Nguyen, 2023)
MBP	2024	GNN	(Yan et al., 2024)
T-ALPHA	2025	Transformer	(Kyro et al., 2025)

The introduction and usage details of the baseline methods ECIF6: LD-GBT, PSH-ML, SybylGGL-Score and T-ALPHA are provided in Section 5.5.1.

Table 3
Hyper-parameter settings of DHIL.

Hyper-parameter	Setting
Batch size	64
d_{atom}, d_{res}, d_{sub}	8 Å, 8 Å, 6 Å
l_{intra}, l_{inter}, l_{atom}, l_{sub}	3, 3, 2, 2
d	128

MBP (Yan et al., 2024) introduces multi-task pre-training, and the prediction of different affinity labels is treated as separate tasks to improve PLBA prediction performance.

To enhance the clarity and completeness of the comparison, representative and widely adopted baseline models from 2020 to 2024, especially the latest GNN-based methods for PLBA predictions, are selected and summarized in Table 2 in chronological order.

5.1.4. Experimental configurations

The DHIL is realized adopting the DGL library (version 1.0.2+cu116) with PyTorch serving as the computational framework (version 1.13.1+cu116). For handling molecular chemical structures, we utilize the RDKit library (version 2022.9.5), ProDy (version 2.4.0), and torchani (version 2.2). The DHIL model is trained on two Tesla P40 GPUs, employing the Adam optimizer with a learning rate of 0.003.

To prevent overfitting, we implement early stopping by monitoring the validation RMSE during training. Specifically, training is halted if no improvement in validation RMSE is observed for 50 consecutive epochs. RMSE is chosen as the early stopping criterion because it is also one of the primary evaluation metrics used during testing, ensuring consistency between model selection and final assessment, and minimizing the risk of selection bias caused by metric mismatch.

Additional hyper-parameter settings are detailed in Table 3.

5.2. DHIL shows the best performance, generalization and robustness

To comprehensively demonstrate the superior performance of DHIL over baseline models in the PLBA prediction task, we first conduct a fair overall performance comparison with baseline methods on the widely used PDBbind *core set*. Subsequently, we explore the generalization capability of DHIL versus baselines on the heterogeneous CSAR-HiQ dataset. Finally, we analyze the robustness of DHIL against suboptimal baselines when faced with interference from low-quality data examples.

5.2.1. Overall performance comparison

To comprehensively evaluate the effectiveness of DHIL in PLBA prediction, we conduct experiments on the widely used PDBbind *core set*. Given that most prior studies (Jiang et al., 2021; Li et al., 2021; Yan et al., 2024; Zhang et al., 2022) primarily rely on the high-quality *refined set* for model training, we first train DHIL and baseline models on this dataset and compare their performance to ensure a fair evaluation. Additionally, we investigate the impact of a larger training dataset by training DHIL on the *general set - refined set* (12,126 samples) and analyzing its performance relative to the model trained on the *refined set*.

Performance of DHIL trained on the refined set and comparison with baselines. We first evaluate the efficacy of DHIL on the PDBbind *core set* and present a comparative analysis of DHIL against baseline methods, as presented in Table 4. It is clear that, DHIL achieves the best performance among all methods, significantly outperforming the second-best baseline. ML-based methods such as GraphDTA and CMPNN primarily rely on SMILES or graph representations to extract covalent ligand information while using sequence-based encoding for proteins. However, their independent modeling of ligands and proteins prevents them from effectively capturing heterogeneous intermolecular interactions, resulting in suboptimal binding affinity predictions. Structure-based approaches such as Pafnucy, OnionNet, SIGN, and SS-GNN focus on learning representations from spatially structured complexes but fail to extract intramolecular interactions within ligands and protein components. Meanwhile, models such as IGN, GLI, and MBP incorporate both intermolecular and intramolecular features but do not thoroughly explore their synergistic relationships, limiting their predictive performance. In contrast, DHIL, with its dual-channel design and hierarchical interaction mechanism, effectively integrates and leverages complementary information from both types of interactions. This enables a more comprehensive modeling of protein-ligand interactions, resulting in superior performance in PLBA prediction compared to all baseline methods.

Analysis of DHIL trained on the general set - refined set. To further investigate the effect of a larger training dataset, we train DHIL on the *general set - refined set* (12,126 samples) and compare its performance with the model trained on the *refined set*. The results, summarized at the bottom of Table 4, indicate that despite a threefold increase in training samples, the performance improvement remains limited. This suggests that dataset quality plays a more critical role than dataset size in PLBA prediction. Several factors contribute to this observation. Compared to the *refined set*, the *general set* contains a greater number of noisy samples, which may introduce biases into the learned features and degrade model performance. Additionally, the *general set* is more prone to data distribution imbalances and redundant samples, negatively impacting the model's generalization ability when evaluated on the high-quality *core set*. Moreover, increasing the dataset size leads to higher optimization complexity, making gradient updates more challenging and potentially affecting model convergence. These findings emphasize that data quality and diversity are more crucial than merely increasing dataset size. The results indicate that the *refined set* alone is sufficient for effective model training and convergence. Therefore, considering the trade-off between computational cost and model performance, we primarily adopt the *refined set* as the main training dataset in this study, with the *general set* serving as a supplementary dataset.

5.2.2. Generalization comparison

Because of the minimal overlap in protein structures between the diverse CSAR-HiQ set and the PDBbind *refined set* (as illustrated in Fig. 3(f)), the models encounter challenges in effectively learning novel target information, significantly impacting their generalization performances. The performance of recent advanced models on the two test sets is summarized in Fig. 4. The figure clearly shows that all baselines exhibit a substantial decrease in performance on the CSAR-HiQ dataset compared to the PDBbind *core set*. This drop underscores the difficulty these models face when applied to data that falls outside their training distribution, revealing a critical limitation in their generalization capabilities. However, DHIL stands out in this comparison by exhibiting a relatively low degradation rate, specifically 22.3 % on RMSE and 2.1 % on R, while maintaining the highest performance on both evaluation metrics. Moreover, in this demanding setting, DHIL makes a greater degree of significant difference (as indicated by the asterisks in Fig. 4) compared to the best baseline model. The statistical significance of these

Table 4
Performance comparison on PDBbind *core set*.

Method		PDBbind *core set*			
		RMSE ↓	MAE ↓	SD ↓	R ↑
ML-based	LR	1.675 (0.000)	1.358 (0.000)	1.612 (0.000)	0.671 (0.000)
	SVR	1.555 (0.000)	1.264 (0.000)	1.493 (0.000)	0.727 (0.000)
	RF-SCore	1.446 (0.008)	1.161 (0.007)	1.335 (0.010)	0.789 (0.003)
CNN-based	Pafnucy	1.585 (0.013)	1.284 (0.021)	1.563 (0.022)	0.695 (0.011)
	OnionNet	1.407 (0.034)	1.078 (0.028)	1.391 (0.038)	0.768 (0.014)
GraphDTA	GCN	1.735 (0.034)	1.343 (0.037)	1.719 (0.027)	0.613 (0.016)
	GAT	1.765 (0.026)	1.354 (0.033)	1.740 (0.027)	0.601 (0.016)
	GIN	1.640 (0.044)	1.261 (0.044)	1.621 (0.036)	0.667 (0.018)
	GAT-GCN	1.562 (0.022)	1.191 (0.016)	1.558 (0.018)	0.697 (0.008)
GNN-based	CMPNN	1.408 (0.028)	1.117 (0.031)	1.399 (0.025)	0.765 (0.009)
	IGN	1.303 (0.052)	1.016 (0.022)	1.300 (0.020)	0.802 (0.033)
	SIGN	1.316 (0.031)	1.027 (0.025)	1.312 (0.035)	0.797 (0.012)
	GLI	1.294 (0.034)	1.026 (0.023)	—	—
	SS-GNN	1.281 (0.021)	1.012 (0.016)	—	0.822 (0.006)
	MBP	1.263 (0.023)**	0.999 (0.024)***	1.229 (0.026)***	0.825 (0.008)**
Ours	DHIL	**1.231 (0.012)**	**0.939 (0.016)**	**1.203 (0.009)**	**0.836 (0.015)**
	DHIL@	1.220 (0.016)	0.935 (0.005)	1.195 (0.023)	0.866 (0.001)

[1]We report the mean and standard deviation of all models, and our DHIL experimental results are based on five random runs.
[2]We conduct two-tailed t-tests to compare DHIL with the second-strongest baseline. The significance of the *P*-values are indicated by ** for less than 0.05 and *** for less than 0.01.
[3]We highlight the top results in bold and the runner-up results with an underline.
[4]The experimental results of ML-based methods, CNN-based methods, and GraphDTA methods are sourced from Li et al. (2021), while the results for GNN-based methods are taken from their respective original papers. — indicates that experimental results are not found in the original paper.
[5]DHIL@ is trained using the *general set - refined set*, whereas the other models are trained exclusively on the *refined set*.

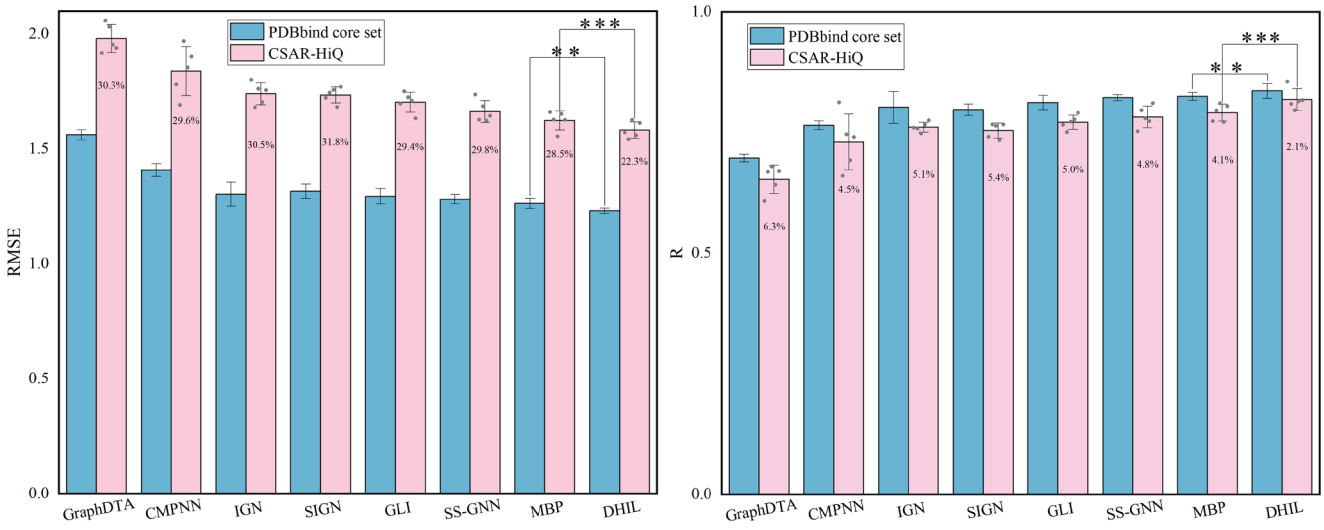


Fig. 4. We report the performance of recent advanced baselines and DHIL trained on the *refined set* and tested on the *core set* and the heterogeneous CSAR-HiQ dataset, evaluated on two metrics. The numbers on the bar charts represent the performance degradation rates of the models, and the error bars indicate standard deviations. We conduct two-tailed t-tests comparing DHIL with the runner-up baseline method. The significance of the *P*-values is denoted by ** for values less than 0.05 and *** for values less than 0.01. The performance metrics of the baseline methods on the *core set* are sourced from the original model publications, while the results on the CSAR-HiQ dataset are derived from five independent random runs of the models provided in their GitHub repositories (including GLI on the PDBbind *core set* under the evaluation metric R).

differences further confirms the reliability of DHIL's predictions on out-of-distribution data.

We further evaluate the generalization ability of DHIL and the runner-up baseline MBP under different protein redundancy rates to assess model performance when reducing sequence similarity between training and test data, ensuring fair validation of overlapping data. Since the CSAR-HiQ dataset and the PDBbind refined set share a certain degree of protein sequence overlap, a strict data filtering strategy is applied to prevent data leakage. Specifically, training samples are selected based on their minimum sequence similarity to the CSAR-HiQ test set, and samples with varying degrees of similarity to the test data are progressively removed to strictly control the overlap between the training and test sets, ensuring fairness in the evaluation and a reasonable assessment of the model's generalization ability. The minimum redundancy rate is set to 0.3, primarily to ensure a sufficient number of training samples for stable model convergence while avoiding excessive data removal that could lead to training instability. Additionally, in the experimental design, only protein sequence similarity is controlled, while

ligand similarity is not considered. This choice is based on the fact that proteins primarily determine the binding pocket and interaction patterns with ligands, making protein similarity adjustment more suitable for evaluating the model's generalization ability to novel targets. Meanwhile, the chemical space of ligands is highly complex, and ligand similarity alone is difficult to use directly as a filtering criterion. Ignoring ligand similarity also helps avoid introducing additional experimental variables that could affect the controllability of the study. As shown in Table 5, the results indicate that as the protein redundancy rate decreases (i.e., proteins in the training set with high similarity to test data are removed), the RMSE of all models increases, suggesting that reducing test-related samples in the training data makes it more challenging for models to learn protein-ligand interactions effectively. However, DHIL consistently outperforms MBP across all redundancy levels, even under the most stringent data removal condition (0.3), where its RMSE remains lower than that of MBP. This demonstrates that DHIL maintains strong predictive performance even under more demanding generalization settings. Further analysis of RMSE variation across different redundancy rates reveals that when the redundancy rate drops to 0.5 or lower, MBP experiences a significant performance degradation (with RMSE increasing by up to 32.76 %), whereas DHIL exhibits a smaller RMSE increase of 27.24 %, consistently maintaining lower RMSE across all data removal settings. This indicates that DHIL is more capable of learning robust and generalizable protein-ligand interaction patterns, reducing reliance on specific training data. In summary, this experiment highlights that DHIL possesses stronger resistance to overfitting, better adapts to unseen protein prediction tasks, and demonstrates more stable performance in out-of-distribution generalization scenarios.

5.2.3. Robustness comparison

Although the accessibility of spatial structure data for complex structures is growing, volume of the high-quality data within *refined set* keeps constrained. In real-world application scenarios, the presence of a large amount of lower-quality data containing noise and incomplete information compels models to possess robustness to enhance practical deployability. To evaluate the model's robustness, we randomly select low-quality samples from the PDBbind *general set* to replace a certain proportion of samples in the *refined set*, aiming to analyze the tolerance of our model and the best baseline when perturbing the training set. As depicted in the Fig. 5, with the increase in replacement proportion, both methods exhibit a declining trend in both metrics. However, DHIL demonstrates relatively stronger robustness, achieving superior performance over the best baseline across all replacement rates. Additionally, as the replacement proportion increases, particularly when the replacement rate exceeds 0.4, the advantage of DHIL over the best baseline expands. Even at high replacement rates, it maintains relatively lower prediction fluctuations, demonstrating the indisputable superior robustness of DHIL.

5.3. Innovative components significantly improve performance

In this subsection, we carry out ablation studies to examine the effectiveness of the novel components in DHIL. Specifically, we extract the intra-type channel encoding, inter-type channel encoding, and dual-scale interactive learning from DHIL to create three variant models. We compile the experimental results of all models on the two datasets in Table 6. Moreover, to assess the impact of each innovative component during training, we plot the convergence curves of two key metrics on the validation set during training on the PDBbind *refined set*, as depicted in Fig. 6.

5.3.1. Effectiveness of dual-channel encoding

As depicted in Table 6, both intra and inter-type channels individually lead to a significant performance decrease of DHIL across various metrics, indicating that modeling only intramolecular or intermolecular interactions is inadequate for each variant to richly grasp the intricate

Table 5
Performance comparison of DHIL and MBP under different protein redundancy rates.

Experimental Details	Model	Protein Redundancy Rate							
		1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.3
RMSE	MBP	1.624 (0.037)	1.630 (0.028)	1.637 (0.063)	1.639 (0.020)	1.645 (0.029)	1.746 (0.057)	1.867 (0.019)	2.156 (0.076)
	DHIL	1.582 (0.032)	1.593 (0.046)	1.620 (0.052)	1.620 (0.032)	1.633 (0.010)	1.738 (0.086)	1.801 (0.059)	2.013 (0.097)
RMSE Reduction Relative to Full Training Set (%)	MBP	–	0.37	0.8	0.92	1.29	7.51	14.96	32.76
	DHIL	–	0.7	2.4	2.4	3.22	9.86	13.84	27.24
Training Sample Count		3767	3723	3696	3653	3573	3231	2934	2500

[1]This table presents the performance comparison between DHIL and the runner-up baseline MBP under different protein redundancy rates. Lower redundancy rates correspond to more stringent filtering of training samples. The selection of training samples is based on the sequence similarity between proteins in the training set and target proteins in the CSAR-HiQ test set. Specifically, for each training sample, the minimum sequence similarity (Li et al., 2020) between its target protein and all proteins in the test set is computed, which is then used as its protein redundancy rate. Training samples with similarity above a predefined threshold are removed.
[2]The RMSE reduction relative to the full training set (redundancy rate of 1.0) quantifies the performance degradation as the available training data decreases. The table also provides the number of training samples at each redundancy rate to illustrate the extent of data removal and further validate the robustness and generalization capability of DHIL under stricter data filtering conditions.

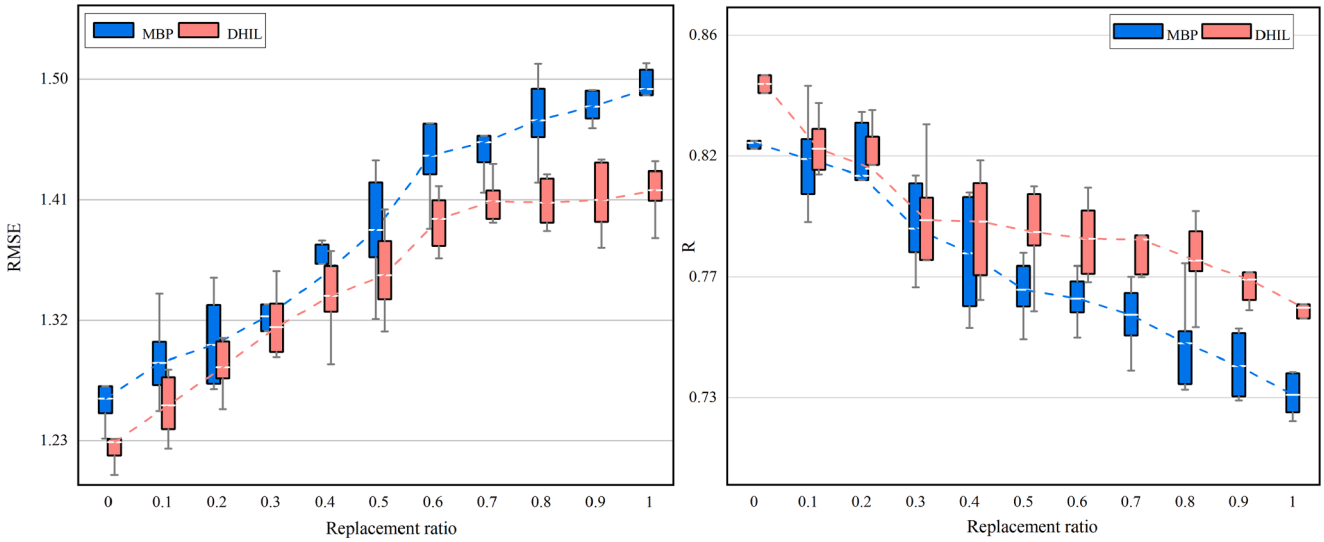


Fig. 5. Boxplots illustrating the performance of DHIL and the second-best baseline on two metrics under different replacement proportions, based on five independent random experiments where low-quality samples from the PDBbind *general set* are used to replace a certain proportion of training samples in the *refined set*. The central line indicates the median value, while the boundaries denote the interquartile range. Whiskers extend to 1.5 times the interquartile range. Additionally, the figure includes dotted lines that signify the mean values.

Table 6
The outcomes of the ablation studies conducted on two testing sets are presented.

Datasets	Models	Intra-type channel	Inter-type channel	Hierarchicalinteractive learning	RMSE ↓	MAE ↓	SD ↓	R ↑
PDBbind	DHIL	✓	✓	✓	**1.231 (0.012)**	**0.939 (0.016)**	**1.203 (0.009)**	**0.836 (0.015)**
	Model-1	–	✓	–	1.513 (0.040)***	1.170 (0.039)***	1.456 (0.032)***	0.735 (0.029)***
	Model-2	✓	–	–	1.476 (0.028)***	1.156 (0.026)***	1.396 (0.023)***	0.807 (0.020)*
	Model-3	✓	✓	–	1.340 (0.020)***	1.135 (0.018)***	1.439 (0.016)***	0.818 (0.020)*
CSAR-HiQ	DHIL	✓	✓	✓	**1.582 (0.032)**	**1.231 (0.035)**	**1.613 (0.013)**	**0.818 (0.020)**
	Model-1	–	✓	–	1.990 (0.056)***	1.484 (0.050)***	1.963 (0.056)***	0.659 (0.049)**
	Model-2	✓	–	–	1.792 (0.043)***	1.296 (0.046)**	1.763 (0.043)***	0.775 (0.033)*
	Model-3	✓	✓	–	1.696 (0.036)***	1.289 (0.040)*	1.691 (0.022)**	0.789 (0.025)*

[1]The average results along with their standard deviations are provided, based on data from five independent random trials.
[2]We conduct two-tailed t-tests to compare DHIL with its variants. The significance of the *P*-values is indicated by *, **, and ***, corresponding to values greater than 0.05, less than 0.05, and less than 0.01, respectively.
[3]We highlight the top results in bold.

molecular interactions involved in the protein-ligand binding process (Model-1 and Model-2). Particularly, relying solely on inter-type channel without considering intramolecular structural information severely limits the model's comprehensive understanding of complex biochemical processes and greatly undermines the accuracy of binding affinity derivations (Model-2). As a comparison, the dual-channel encoding design for DHIL explicitly captures both intramolecular and intermolecular interactions, thereby capturing more comprehensive interaction features and significantly enhancing performance.

Additionally, as observed from Fig. 6, the rich encoding features of the dual-channel architecture enable DHIL to maintain the most rapid and stable learning during training. Nevertheless, solely encoding intramolecular covalent information (Model-1) without incorporating distance guidance between diverse atoms renders the model susceptible to local overfitting when extracting essential affinity features. This, in turn, exacerbates fluctuations during the training process. Hence, the dual-channel encoding provides comprehensive and highly correlated interaction information for DHIL, demonstrating an effective and efficient capability to encode features of complex biological interaction systems.

To showcase the efficacy of the dual-channel feature encoding of DHIL, we render the clustering patterns for the embedded vectors for ligands and protein atoms across 290 complexes from the PDBbind *core set*. We apply t-distributed stochastic neighbor embedding (t-SNE) (Van der Maaten & Hinton, 2008) to lower the dimensionality of the embedded vectors, specifically those from the intra and inter-type channels in the final iteration, mapping them into a two-dimensional space for clearer visualization. As depicted in Fig. 7, the embedded ligand and protein atoms learned from the intra-type channel exhibit a compact, mixed distribution of multiple clusters, reflecting the close connectivity of similar internal structures due to the small distances between atoms connected by covalent bonds. However, after the atoms of both types undergo updates through the inter-type channel, the embedded atoms of the two types exhibit two distinct dispersion patterns, reflecting heterogeneous interaction relationships between the two types of atoms mediated by long-range noncovalent interactions. These visualization results support the ability of our dual-channel encoding to distinguish between different types of interactions.

5.3.2. *Effectiveness of hierarchical interactive learning*

To validate the effectiveness of hierarchical interactive learning, we replace the original interactive paradigm in DHIL with a simple vector concatenation to fuse intra and inter-type features, represented as Model-3. From Table 6, the performance of Model-3 shows a decline on both datasets compared with DHIL, especially in the RMSE metric. This is because simple concatenation of the two-channel features is insufficient to capture the synergies between them, whereas DHIL, through the elaborately designed dual-scale interactive learning, achieves bidirectional information transfer between the two channels, which allows

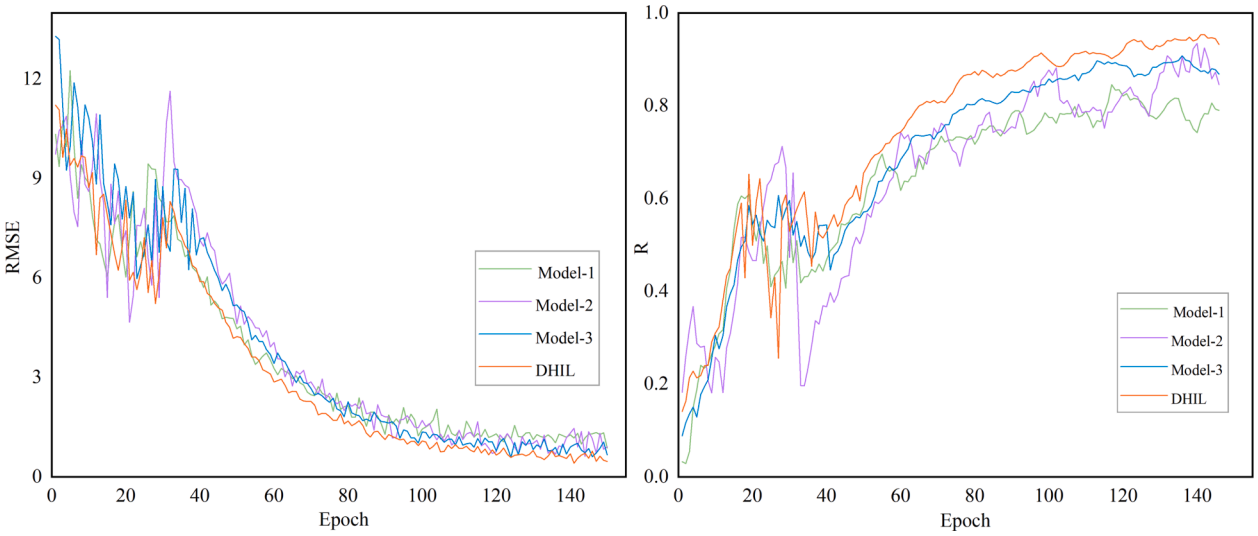


Fig. 6. The plot shows convergence curves of two key metrics on the validation set during training on the PDBbind *refined set* when removing key components.

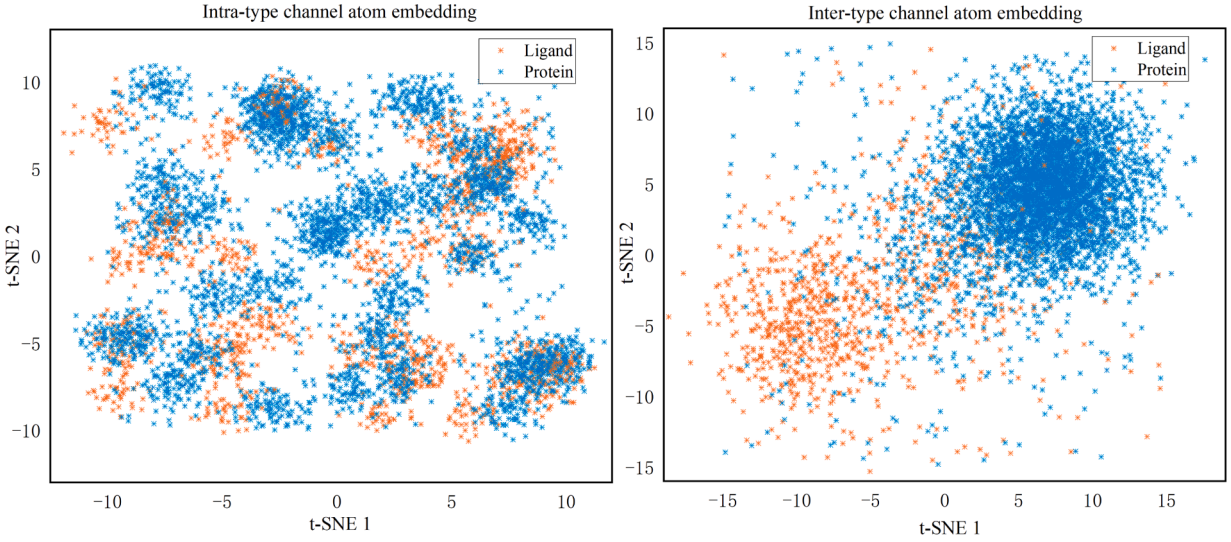


Fig. 7. The figure illustrates the clustering patterns of the final embedded vectors for ligand and protein atoms from 290 complexes in the PDBbind *core set*, with vectors originating from the intra and inter-type channels, respectively. These vectors are mapped into a two-dimensional space using t-SNE, from their original d-dimensional representation.

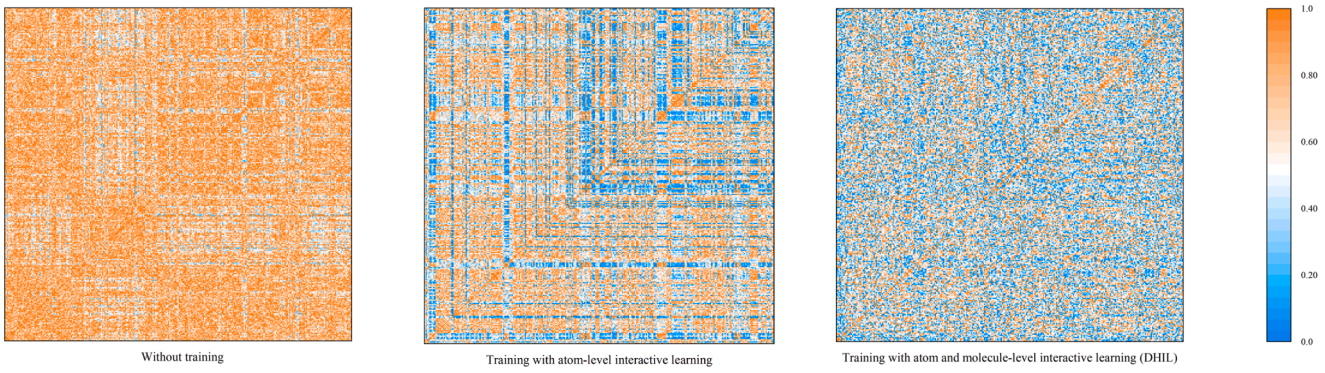


Fig. 8. The heat map illustrates the similarity matrix for 290 samples in the *core set* using DHIL and its variants. It compares three scenarios: (a) without any training, (b) with training that involves only atom-level interactive learning, and (c) with training that includes both atom-level and substructure-level interactive learning.

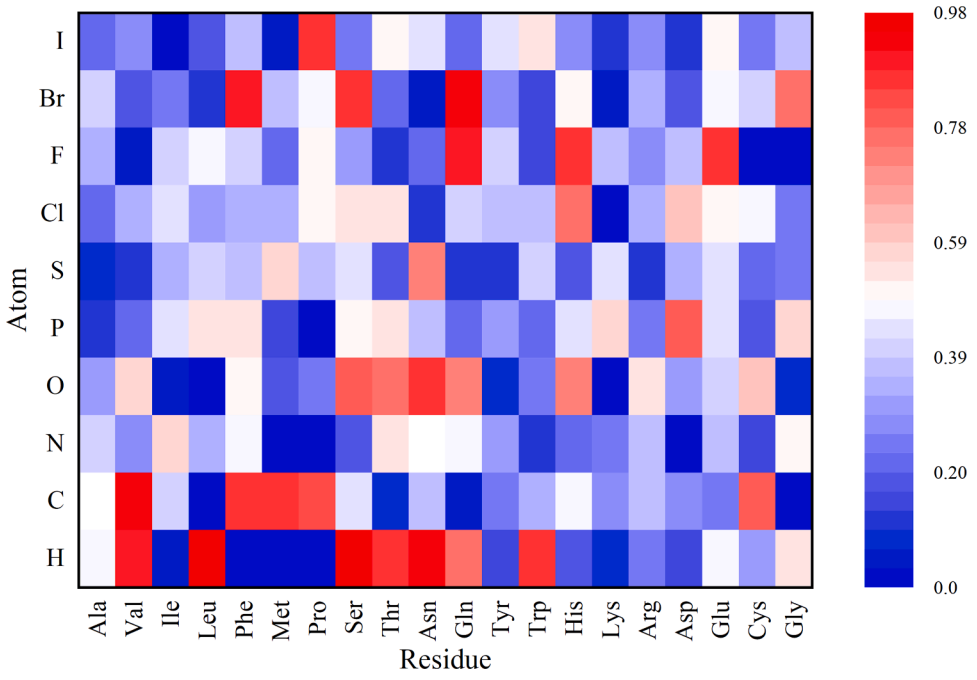


Fig. 9. The heatmap matrix depicts the interaction strengths between 20 prevalent residues and 10 atom types across 290 complexes within the *core set*, as inferred by the model.

for a more comprehensive modeling of the collaborative correlations between the two types of interactions, resulting in superior performance and the faster convergence (DHIL compared with Model-3 in Fig. 6).

To furtherly illustrate the role of information interaction at both the atom and substructure levels in hierarchical interactive learning, Fig. 8 visualizes the initial and interactively learned vectors decoded by the layer of the weight (derived from Eq. (29)) for 290 complexes in the PDBbind *core set* using heatmaps. The representation vectors are compared using the Pearson's correlation coefficient (R). A larger R value indicates a higher similarity between the two vectors, and vice versa. From the figure, before training, the model demonstrates high similarity scores, suggesting challenges in discerning between diverse complexes. However, after training, there is a noticeable shift: from atom-level interactive learning to substructure-level ones, the model reveals an escalating complexity in the similarity matrix. This suggests an enhanced ability to capture nuanced interaction patterns, thereby improving its performance. These findings are consistent with established domain knowledge (Brown et al., 2013; Chen & Kurgan, 2009), affirming the hierarchical interactive learning's efficacy in capturing intricate molecular interactions.

5.4. DHIL exhibits strong biochemical interpretability

In the previous sections, we have demonstrated the strong performance of DHIL in affinity prediction tasks. This can be attributed to the model's capacity to learn the molecular mechanisms underlying protein-ligand interactions. To further enhance DHIL's interpretability and gain insights into the relevant interaction theories in modeling interactions, we investigate the correlation between the learned representations of the atoms and bridge nodes and the physicochemical patterns of protein-ligand binding. Through the analysis of these correlations, our objective is to clarify how DHIL can deliver valuable understanding of the fundamental mechanisms governing protein-ligand interactions.

5.4.1. DHIL identifies key atoms and residues constituting binding sites

The binding between ligands and proteins is primarily achieved through hydrogen bonds, hydrophobic interactions, van der Waals forces, and electrostatic interactions formed between ligand atoms and protein residues. Therefore, in affinity prediction tasks, analyzing the ability of DHIL to recognize the atoms and residues at the binding site is crucial for assessing the biochemical theoretical support in the model's decision-making process. Examining the similarity between the final atom representations (or substructure representations) learned by DHIL reveals the model's capability to autonomously learn binding sites. Notably, for a more discernible visualization, we opt to show atom-residue interaction weights rather than atom-atom interaction weights. We first compute the similarity value between two atoms by calculating the Pearson's correlation coefficient (R) between their final representations derived from $\mathbf{Z}_I^{atom^{**}}$. Then, we employ a computational method (from Jiang et al. (2021)) to calculate the weight for each atom-residue pair by averaging the similarity values between all protein atoms in the involved residue and the ligand atom in the atom-residue pair. Similarly, the importance of a ligand atom is determined by averaging its similarity value with each protein atom. We conduct a analysis of the interaction weights between 20 prevalent residues and 10 types of atoms across 290 complexes from the *core* set, and depict these weights as a matrix heatmap in Fig. 9.

Through the heatmap visualization of the interaction matrix, we can observe high-frequency interactions between specific residues and atoms. For example, residues such as serine (Ser), threonine (Thr), asparagine (Asn), and glutamine (Gln) exhibit strong interaction intensities with hydrogen (H) and oxygen (O) atoms, indicating that these residues play a crucial role in forming hydrogen bonds. Additionally, hydrophobic residues like valine (Val), and phenylalanine (Phe), methionine (Met), and proline (Pro) show significant interactions with carbon (C) atoms, which reflects the importance of hydrophobic interactions in protein-ligand binding. Furthermore, the high interaction intensities between bromine (Br) atom with certain residues, such as Phe, Ser, Gln, and luorine (F) with Gln, histidine (His), and glutamic acid (Glu), may suggest the strong interactions between the polar groups (e.g., hydroxyl, amide groups) in these amino acid side chains and bromine and fluorine. These observations validate common biochemical interaction mechanisms and demonstrate that the DHIL model has a certain degree of accuracy in recognizing these key binding sites.

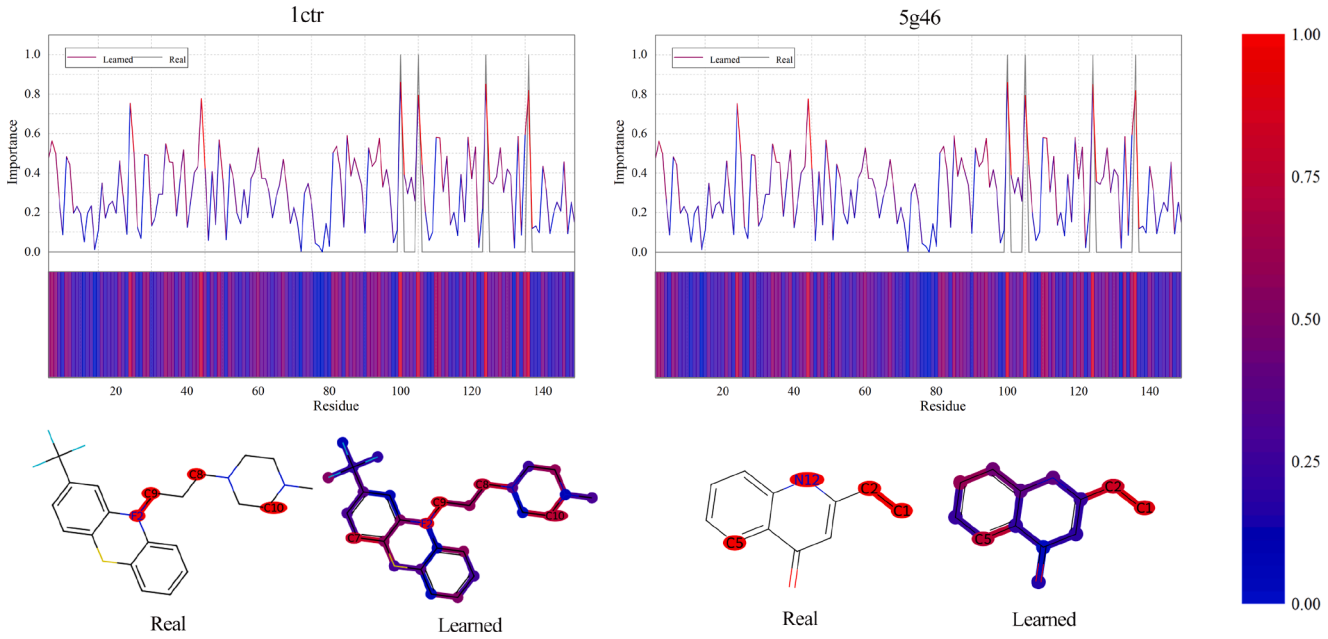


Fig. 10. Visualization of real and learned interaction importance for residues and atoms in complexes 1ctr and 5g46. The upper line charts (real and learned) and heatmaps (predicted) display the importance of residues in the 1ctr and 5g46 complexes. The lower panels show the real and learned atom weight distributions for the ligand structures.

To concretely demonstrate DHIL's ability to perceive binding sites, as shown in Fig. 10, we select two cases (PDB IDs: 1ctr and 5g46) and visualize the real and predicted interaction importance of their residues and atoms. The binding site labels in the complexes are identified using the PLIP tool (Salentin et al., 2015). Protein residues are aligned with the UniProt sequence utilizing sequence alignment utilities (Zhao et al., 2013), while ligand atoms are correlated with their respective structures (acquired from http://ligand-expo.rcsb.org/ld-download.html), guaranteeing distinct naming and indexing for all non-hydrogen atoms. In the case of 1ctr, the polylines illustrate the uniformity of the peak intensities, with the heatmap's high-intensity zones signifying the model's proficiency in identifying the critical residues that partake in interactions. Moreover, the paralleling visualization of the actual and forecasted structures of the interacting ligands at the base of Fig. 10 underscores DHIL's capacity to pinpoint pivotal atoms. For 5g46, due to its more complex spatial structure (518 residues) and the extremely imbalanced ratio of positive to negative samples (4:514), the model's learning results for key residues and atoms show more noticeable deviations compared to 1ctr. However, DHIL still demonstrates a certain ability to recognize crucial information under such challenging scenarios.

In brief, despite the absence of any annotated site information for supervision during the derivation of protein-ligand binding affinity, DHIL demonstrates a decent ability to recognize the importance of atoms and residues at binding sites. This capability enhances the interpretability of the model, providing reliable guidance for subsequent wet-lab experimental validation.

5.4.2. The local-to-global bridge node setting simulates the operation of biological systems from local to global levels

To intuitively analyze the rationality and effectiveness of the hierarchical bridge node setup in feature extraction and representation of protein-ligand interactions, we employ t-SNE to visualize the dimensionality reduction of the final representations of bridge nodes across all test samples in the PDBbind *core set*. As shown in Fig. 11(a), the bridge nodes exhibit an approximate two-cluster distribution, corresponding to protein and ligand clusters. This indicates that the bridge nodes effectively capture the multi-type representations of the components in the complexes.

Moreover, to investigate the distribution patterns of local and global bridge nodes, we highlight the local and global bridge nodes involved in single sample. We carefully select five cases, as depicted in Fig. 11((b)-(f)). We find that higher-level nodes (such as global bridge nodes of proteins or ligands) are located at the center of the distribution, whereas lower-level nodes (such as local bridge nodes of proteins or ligands) surround them. This distribution pattern demonstrates that our model successfully captures the hierarchical relationships within the protein and ligand structures. Global bridge nodes are responsible for aggregating and distributing overall structural information, ensuring the model can identify the global matching of substructures between proteins and ligands. Local bridge nodes, on the other hand, convey specific local structural information, including the arrangement and interactions of individual atoms. These local details are crucial for refining the precise interactions at the binding sites between proteins and ligands. The hierarchical information transfer mechanism facilitated by bridge nodes enables the model to achieve a detailed and comprehensive understanding of the complex interactions between proteins and ligands, which allows for the coordinated operation of structural and functional units at each level, functioning as an integrated whole.

Based on the aforementioned analysis of the roles of bridge nodes, we aim to ascertain whether the sequence of setting local and global bridge nodes affects model performance, i.e., whether substructure-level information transmission should precede atom-level information transmission. To this end, while keeping other model settings constant, we conduct an additional experiment (Model-4) by only altering the sequence of the two levels of information transmission. The results of this experiment are summarized in Table 7. From the table, it can be observed that the model with atom-level information transmission first, followed by substructure-level information transmission, outperforms the Model-4 with the reverse sequence across all evaluation metrics on both datasets. We believe that processing atom-level information first helps DHIL optimize these critical microscopic interactions at the initial stage, providing a solid foundation for subsequent global information integration. Conversely, starting with substructure information transmission may overlook or obscure these important fine-grained details,

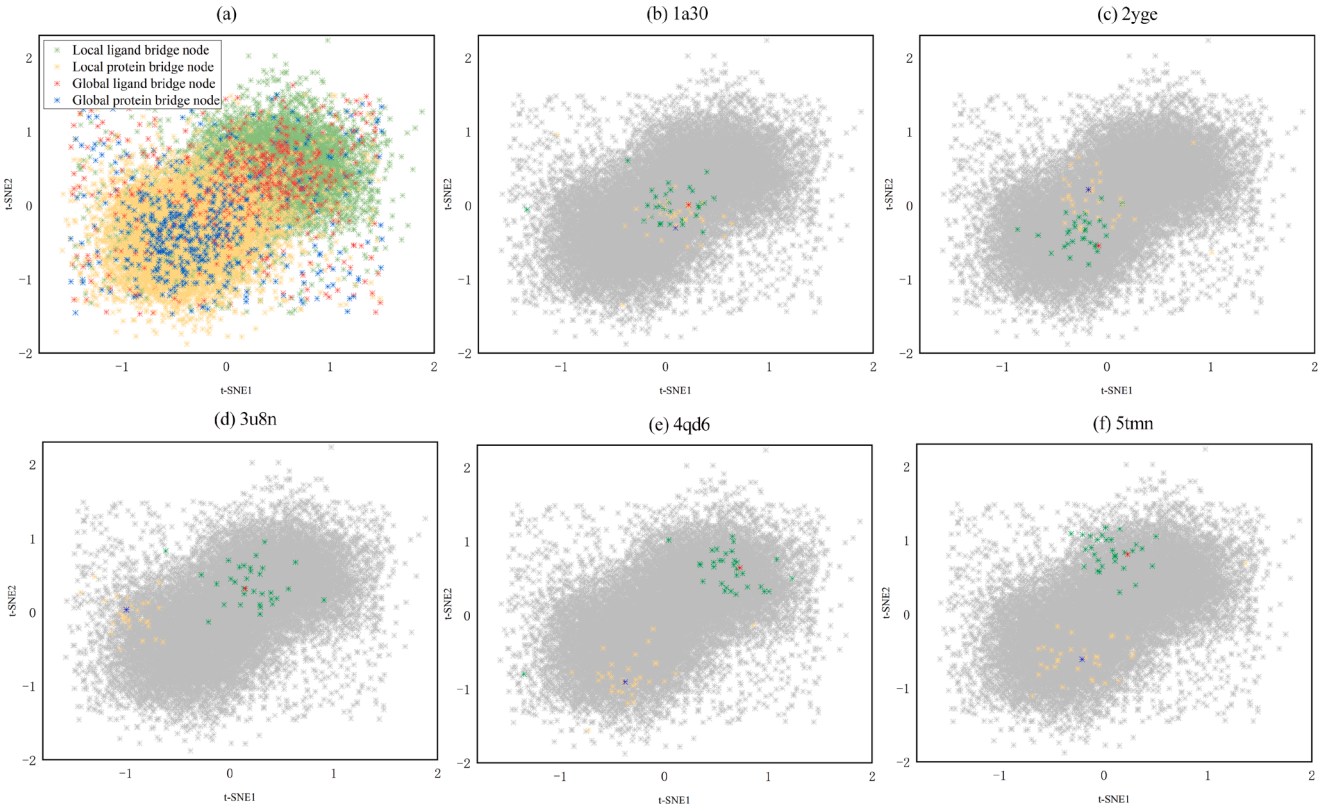


Fig. 11. The final representations of bridge nodes involved in all test samples of the PDBbind *core set* are visualized in two dimensions using t-SNE. (a) The dimensionality reduction distribution of all bridge nodes in all test samples. (b)-(f) The dimensionality reduction highlighting the bridge nodes involved in five individual samples (PDB IDs: 1a30, 2yge, 3u8n, 4qd6, and 5tmn).

Table 7
The performance of DHIL and its variants.

Method	PDBbind *core set*				CSAR-HiQ set			
	RMSE ↓	MAE ↓	SD ↓	R ↑	RMSE ↓	MAE ↓	SD ↓	R ↑
DHIL	**1.231 (0.012)**	**0.939 (0.016)**	**1.203 (0.009)**	**0.836 (0.015)**	**1.582 (0.032)**	**1.231 (0.035)**	**1.613 (0.013)**	**0.818 (0.020)**
Model-4	1.263 (0.018)	0.958 (0.026)	1.236 (0.016)	0.815 (0.009)	1.610 (0.029)	1.264 (0.023)	1.635 (0.006)	0.779 (0.019)
Model-5	1.308 (0.031)	1.026 (0.026)	1.319 (0.040)	0.800 (0.006)	1.745 (0.052)	1.332 (0.054)	1.720 (0.049)	0.749 (0.102)
Model-6	1.398 (0.040)	1.125 (0.049)	1.438 (0.037)	0.721 (0.001)	1.802 (0.060)	1.420 (0.065)	1.817 (0.047)	0.723 (0.009)

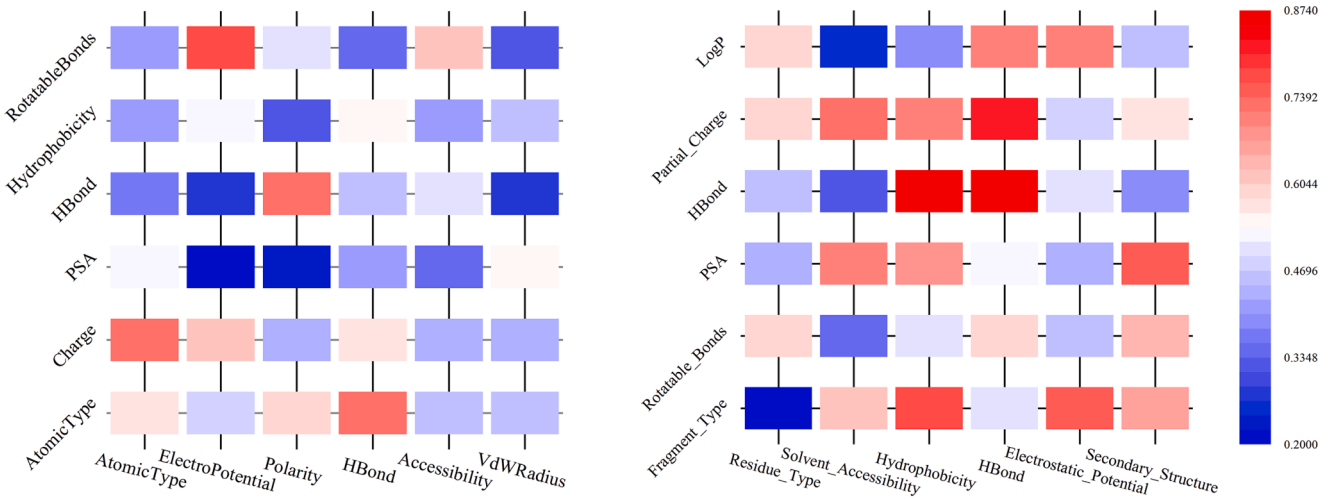
[1]The performance of DHIL and its variants, including Model-4, which alters the order of hierarchical information transfer, as well as Model-5 and Model-6, which replace the residue-level protein substructure with secondary structures and functional domains, respectively, on the PDBbind *core set* and CSAR-HiQ set.
[2]The average results along with their standard deviations are provided, based on data from five independent random trials.
[3]We highlight the top results in bold.

thus impacting overall performance. More importantly, the sequence of local-first-then-global (atom-level followed by substructure-level) information transmission aligns with the information transmission mechanisms in biological systems, which enables the model to more naturally emulate the functioning of biological systems.

In summary, the local-to-global bridge node setting not only effectively captures the protein and ligand components within the complexes but also naturally simulates the flow of information from local to global levels as observed in biological systems. This demonstrates that DHIL possesses strong model interpretability and biological relevance.

5.4.3. The role of hierarchical node features in protein-ligand interaction modeling

In the DHIL model, we employ a two-level feature modeling approach, incorporating atomic and substructural features to learn protein-ligand interactions, ensuring that the model captures both local microscopic interactions and global binding patterns. Atom-level features (such as electrostatic potential, hydrogen bond donors and acceptors, and hydrophobicity) determine the strength and stability of local interactions, while substructure-level features (such as aromatic rings, polar groups, and secondary structures) influence more macroscopic binding patterns, including the formation of hydrophobic cores and secondary structure adaptability. DHIL utilizes a dual-channel architecture to integrate interactive information across different scales, enabling the model to comprehensively consider the contributions of these two feature types to affinity prediction. To further validate the biochemical plausibility of the model, we constructed a feature importance heatmap at both atomic and substructural levels (Fig. 12), analyzing DHIL's attention focus when learning interactions at different scales. This analysis not only enhances the credibility of DHIL's predictive re-



(a) Atomic feature importance heatmap

(b) Substructure feature importance heatmap

Fig. 12. Atomic and substructure feature importance heatmaps for protein-ligand interaction modeling in DHIL. (a) The atomic feature importance heatmap represents the pairwise interactions between different physicochemical properties of protein and ligand atoms and their impact on affinity prediction. (b) The substructure feature importance heatmap illustrates the pairwise interactions between different features of protein residues and ligand fragments and their contribution to predictive performance. Feature importance is evaluated by systematically removing pairs of features and measuring their impact on model prediction error (RMSE), followed by Z-score normalization.

sults but also deepens the understanding of the model's learning mechanisms in protein-ligand affinity prediction, providing valuable guidance for future network architecture improvements and feature design optimizations.

From the atomic feature importance heatmap Fig. 12(a), it can be observed that electrostatic interactions exhibit the highest importance, which aligns well with the DHIL atom-level scale, as this level focuses on modeling local interactions. The electrostatic potential distribution of protein atoms directly determines the binding pattern of charged ligands, and DHIL enhances the learning of electrostatic interactions at this scale through GNNs, thus significantly highlighting the importance of this feature. Meanwhile, hydrogen bonding also shows relatively high importance, indicating that DHIL effectively captures the role of the hydrogen bond network in binding stability and learns the interaction patterns between hydrogen bond donors and acceptors through its dual-channel interactive mechanism. In contrast, hydrophobic interactions have a lower impact, possibly because DHIL is designed to prioritize charged and polar interactions, making it less dependent on hydrophobic interactions. This could reflect that the protein binding pockets in the dataset predominantly feature polar environments, or that DHIL has already incorporated relevant hydrophobic information at the substructural level, thereby reducing reliance on the atom-scale representation. In the substructural feature importance heatmap Fig. 12(b), DHIL captures the global characteristics of the binding site through larger-scale structural patterns, particularly highlighting the high importance of protein secondary structures, ligand fragment types, and hydrogen bond donor/acceptor features. This suggests that at the substructural scale, DHIL effectively learns how the three-dimensional arrangement of protein residues influences ligand binding while reinforcing the overall hydrogen bonding pattern of the binding pocket. These results indicate that DHIL's substructural-scale modeling does not solely rely on local atomic interactions but also integrates broader molecular fragment information, thereby improving prediction stability and generalization capability. Additionally, ligand polar surface area and protein solvent accessibility exhibit relatively high importance, suggesting that DHIL effectively learns the impact of polarity matching on binding affinity at this scale. The significance of these features further indicates that DHIL can effectively distinguish the polarity of binding pockets in its global structural modeling process, thereby optimizing the assessment of ligand adaptability to the binding site.

Overall, DHIL, through its dual-channel hierarchical architecture, effectively integrates information across different scales, forming a comprehensive protein-ligand binding modeling framework that spans from microscopic atom-level interactions (e.g., electrostatics, hydrogen bonding) to macroscopic substructural adaptations (e.g., secondary structures, solvent accessibility). The atomic feature importance heatmap reflects DHIL's capability in learning electrostatic potential at the lower-level channel, while the substructural feature importance heatmap demonstrates that DHIL enhances its understanding of protein-ligand compatibility through global structural information.

5.4.4. Residue-level features at the substructure scale better capture protein-ligand binding patterns

To further investigate the impact of protein substructure granularity on PLBA prediction performance, Model-5 and Model-6 are constructed as experimental variants. In Model-5, secondary structures replace the residue-level substructures, while in Model-6, functional domains serve as the protein substructures. In constructing Model-5, protein PDB structure files are extracted from the PDBbind dataset, and the DSSP (Define Secondary Structure of Proteins) tool is employed to annotate the secondary structure information. Based on DSSP's classification of -helices, -sheets, and random coils, continuous secondary structure units are aggregated into protein substructures. For Model-6, functional domain information is primarily obtained from the Pfam database; in cases where pre-annotated domains are unavailable, HMMER is used for sequence-based domain prediction, supplemented with three-dimensional structural analysis to ensure the accuracy and integrity of functional domain segmentation. In both Model-5 and Model-6, only the protein substructure representation is modified, while the ligand substructure representation remains unchanged. The newly constructed protein substructure features are then embedded into the DHIL model and trained using the same dataset and hyperparameters to compare the effects of different protein substructure granularities on model performance.

The experimental results from Table 7 demonstrate that the DHIL model achieves the best performance on both the PDBbind core set and CSAR-HiQ set, exhibiting the lowest RMSE and MAE as well as the highest R. This indicates that using residue-level protein substruc-

tures enables more effective learning of protein-ligand interaction features, thereby enhancing the accuracy of binding affinity predictions. In contrast, the performance of Model-5 (secondary structure-based) and Model-6 (functional domain-based) declines across all evaluation metrics. This decline is particularly pronounced on the CSAR-HiQ dataset, where both RMSE and MAE are significantly higher than those of DHIL, suggesting that these coarser protein substructure representations lead to information loss, thereby reducing the model's generalization ability. Specifically, Model-5 (secondary structure-based) exhibits increased RMSE and MAE compared to DHIL. For instance, on the PDBbind core set, RMSE rises from 1.231 to 1.308, and R decreases from 0.836 to 0.800, indicating that modeling protein substructures at the secondary structure level may overlook key residues within binding sites. Since secondary structures typically encompass multiple amino acid residues, critical binding residues may be diluted by irrelevant structural components, making it more difficult for the model to accurately capture essential interactions. Moreover, on the CSAR-HiQ dataset, RMSE increases from 1.582 to 1.745, further indicating that secondary structure segmentation hinders the model's generalization to diverse protein-ligand systems. Model-6 (functional domain-based) performs the worst across all evaluation metrics, with both RMSE and MAE significantly higher than those of DHIL and Model-5. For example, on the PDBbind core set, RMSE increases to 1.398, and R drops to 0.721; on the CSAR-HiQ set, RMSE reaches 1.802, and R falls to 0.723. This suggests that in binding affinity predictions, using functional domains as protein substructure units is overly coarse, making it difficult to precisely capture protein-ligand interaction details. Functional domains often span large structural regions of proteins, some of which may not be directly involved in ligand binding, leading to the inclusion of excessive irrelevant information during model training and ultimately degrading prediction accuracy.

Overall, the experimental results indicate that residue-level protein substructure representation remains the optimal choice for PLBA predictions, while secondary structure- and functional domain-level representations compromise model performance to some extent. Although secondary structure-based representation retains some local structural information, it lacks sufficient granularity to accurately capture fine-scale details at critical binding sites. On the other hand, functional domain-based representation, due to its broader coverage, introduces excessive irrelevant information, further diminishing model performance.

5.5. Extended validation on additional datasets

To further evaluate the performance of the DHIL model, this section introduces additional benchmark datasets for experimental analysis. First, the latest version of the PDBbind dataset is used to assess the impact of dataset updates on the model's predictive capability, while comparisons with multiple baseline methods ensure a comprehensive evaluation. Second, virtual screening experiments are conducted using the DUD-E dataset to validate DHIL's screening ability in drug discovery, particularly in identifying active compounds.

5.5.1. Performance validation on the latest PDBbind v2020

Compared to v2016, PDBbind v2020 expands the dataset size, increasing the number of protein-ligand complexes from approximately 17,340 to 19,443, while the refined set extends to 5316 high-quality complexes. Moreover, PDBbind v2020 exhibits greater diversity in data types and undergoes stricter filtering and quality control to improve data reliability. The inclusion of new data covers more underexplored protein families and integrates advanced structural resolution techniques, enhancing the model's generalization ability. By training and evaluating DHIL and baseline methods on PDBbind v2020, a comprehensive analysis of the impact of dataset updates on model prediction performance is conducted.

In terms of baseline model selection, we include four additional models—PSH-ML (Liu et al., 2021), ECIF6: LD-GBT (Sánchez-Cruz et al., 2021), SybylGGL-Score (Rana & Nguyen, 2023), and T-ALPHA (Kyro et al., 2025)—on top of the suboptimal model MBP, to ensure the comprehensiveness and timeliness of the experimental evaluation. PSH-ML adopts a spectral graph learning approach based on high-order topological features and shows favorable performance in protein-ligand binding affinity prediction tasks. ECIF6: LD-GBT integrates Extended-Connectivity Interaction Features (ECIF) with Graph Neural Networks (GNNs) to better capture protein-ligand interactions. SybylGGL-Score combines molecular force fields with GNNs, achieving complementary modeling between traditional scoring functions and deep learning architectures. T-ALPHA uses a Transformer-based hierarchical deep learning framework with multi-scale attention mechanisms to effectively model protein-ligand binding patterns. These models not only enrich the performance evaluation but also verify the generalization ability of DHIL across different feature spaces, model structures, and learning strategies, thereby enhancing the comparability and credibility of the experimental results.

As shown in Table 8, all models train on two types of training sets: the general set minus the refined set (larger in size but lower in data quality) and the refined set (consisting of high-quality complexes). Testing is conducted on the PDBbind v2016 core set, since PDBbind v2020 does not provide an updated core set for evaluation. Results show that although the general set minus the refined set is approximately 2.7 times larger than the refined set, the performance improvement in RMSE, MAE, and R remains limited. For example, DHIL achieves an RMSE of 1.216 and an MAE of 1.053 when trained on the General set minus the refined set, while the refined set yields 1.233 and 1.084, corresponding to only 1.4 % and 2.9 % improvement, respectively. This suggests that simply increasing the amount of low-quality data does not lead to substantial gains in model performance. In contrast, when comparing models trained on PDBbind v2020 with those trained on v2016, all methods exhibit more significant improvements in predictive performance. Specifically, across the six models, the average RMSE decreases by 8.21 %, MAE by 4.47 %, SD by 4.19 %, and R increases by 3.28 %. These results underscore the advantage of PDBbind v2020 in providing a larger and higher-quality dataset, which facilitates more effective model training and better captures the complexity of protein-ligand interactions. More importantly, under all experimental configurations—regardless of the scale or quality of training data—DHIL consistently outperforms all baseline models in terms of RMSE, MAE, SD, and R, demonstrating superior predictive accuracy and robustness.

5.5.2. Virtual screening validation on DUD-E

In real-world drug discovery, virtual screening (VS) serves as an essential technique for efficiently identifying potential active compounds. However, traditional protein-ligand binding affinity prediction models primarily focus on the binding strength of known complexes, lacking an assessment of their screening capabilities across large chemical spaces. To evaluate the model's effectiveness in practical screening tasks, this subsection incorporates virtual screening experiments and utilizes the Directory of Useful Decoys - Enhanced (DUD-E (Mysinger et al., 2012)) dataset as a benchmark. DUD-E is one of the most widely used benchmark datasets in virtual screening research, covering 102 distinct protein targets across multiple key categories, including kinases, nuclear receptors, G protein-coupled receptors (GPCRs), proteases, and ion channels. The dataset contains 22,805 experimentally determined active compounds, along with 1,411,214 decoy molecules that share similar physicochemical properties with the active ligands but are theoretically non-binding to the target proteins. This enables DUD-E to effectively evaluate a model's ability to distinguish true active ligands from non-specific binding molecules, providing a reliable measure of its practical value in virtual screening tasks. In the experiments, to ensure the representativeness of the test set, six targets are randomly selected from each of the five major protein categories (kinases, nuclear receptors, GPCRs, proteases, and ion channels), resulting in a total of 30 test targets, while the remaining targets are used for training and validation.

Table 8
Performance comparison of DHIL and baseline models on PDBbind v2020.

Method	Training Set	Test Set	RMSE ↓	MAE ↓	SD ↓	R ↑
PSH-ML	*general set - refined set*	PDBbind v.2016 *core set*	1.300 (0.032)	1.250 (0.021)	1.200 (0.006)	0.835 (0.004)
ECIF6: LD-GBT	*general set - refined set*	PDBbind v.2016 *core set*	1.270 (0.041)	1.119 (0.036)	1.190 (0.007)	0.858 (0.002)
sybylGGL-Score	*general set - refined set*	PDBbind v.2016 *core set*	1.231 (0.041)	1.083 (0.034)	1.170 (0.006)	0.870 (0.003)
MBP	*general set - refined set*	PDBbind v.2016 *core set*	1.225 (0.035)	1.067 (0.033)	1.150 (0.006)	0.872 (0.002)
T-ALPHA	*general set - refined set*	PDBbind v.2016 *core set*	1.235 (0.035)	1.086 (0.023)	1.173 (0.001)	0.867 (0.002)
DHIL	*general set - refined set*	PDBbind v.2016 *core set*	**1.216 (0.036)**	**1.053 (0.025)**	**1.130 (0.005)**	**0.879 (0.004)**
PSH-ML	*refined set*	PDBbind v.2016 *core set*	1.460 (0.025)	1.306 (0.022)	1.302 (0.005)	0.798 (0.004)
ECIF6: LD-GBT	*refined set*	PDBbind v.2016 *core set*	1.310 (0.022)	1.167 (0.021)	1.253 (0.003)	0.830 (0.004)
sybylGGL-Score	*refined set*	PDBbind v.2016 *core set*	1.260 (0.023)	1.123 (0.024)	1.246 (0.004)	0.844 (0.005)
MBP	*refined set*	PDBbind v.2016 *core set*	1.258 (0.039)	1.099 (0.035)	1.240 (0.001)	0.851 (0.005)
T-ALPHA	*refined set*	PDBbind v.2016 *core set*	1.263 (0.049)	1.125 (0.032)	1.250 (0.012)	0.843 (0.007)
DHIL	*refined set*	PDBbind v.2016 *core set*	**1.233 (0.030)**	**1.084 (0.020)**	**1.236 (0.001)**	**0.860 (0.003)**

[1]The average results along with their standard deviations are provided, based on data from five independent random trials.
[2] We emphasize the top results in bold and the suboptimal results with underlines.

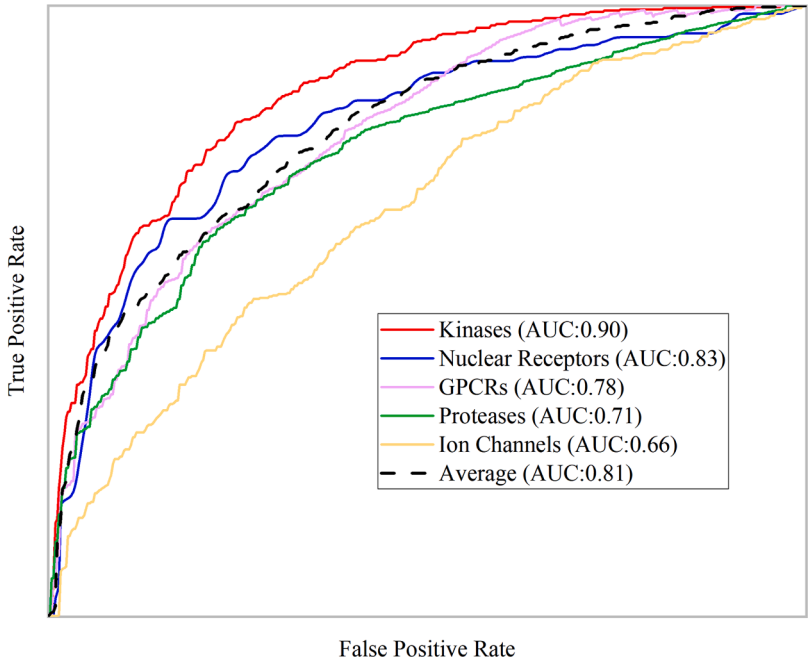


Fig. 13. The average ROC curves of DHIL on the DUD-E dataset for five different protein types: kinases, nuclear receptors, G protein-coupled receptors (GPCRs), proteases, and ion channels. The x-axis represents the false positive rate (FPR), indicating the proportion of non-active compounds misclassified as active, while the y-axis represents the true positive rate (TPR), showing the proportion of correctly identified active compounds. Each solid line corresponds to the average ROC curve for a specific protein type, with the area under the curve (AUC) values provided in the legend. The dashed black line represents the overall average performance across all protein types, demonstrating the general virtual screening capability of DHIL.

To adapt DHIL for the virtual screening task, the loss function is modified by replacing the regression loss with binary cross-entropy (BCE) loss. Since the original DHIL task focuses on regression-based binding affinity prediction, whereas virtual screening is a binary classification problem (active ligand vs. decoy molecule), the model no longer directly optimizes continuous affinity scores. Instead, it learns to establish a classification boundary between active ligands and decoys. In this approach, the model's output values are normalized using a Sigmoid function, mapping predictions to the [0,1] range, where the output can be interpreted as the probability of ligand activity. During training, DUD-E dataset labels (1 for active ligands, 0 for decoys) serve as supervision signals, and binary cross-entropy loss is optimized to minimize classification errors, ensuring that the model effectively distinguishes active ligands from non-binding decoy molecules.

ROC-AUC (Receiver Operating Characteristic - Area Under the Curve) is used as the screening performance metric. As shown in Fig. 13, the overall average AUC across 30 protein targets reaches 0.81, indicating that DHIL exhibits strong virtual screening capabilities. From the perspective of different protein categories, kinases achieve the highest AUC at 0.90, suggesting that DHIL effectively captures kinase active pocket features and models ligand binding patterns with high accuracy. Kinases typically have well-defined ATP-binding pockets, and their ligand interaction patterns tend to be more consistent, enabling DHIL's hierarchical interaction learning to fully utilize structural information and enhance predictive performance. In contrast, nuclear receptors and G protein-coupled receptors (GPCRs) show slightly lower AUCs of 0.83 and 0.78, respectively. Nuclear receptors generally have more open binding pockets that accommodate diverse ligand types, while GPCRs exhibit significant conformational variability, introducing structural uncertainty and making precise modeling more challenging. Proteases achieve an AUC of 0.71, whereas ion channels have the lowest AUC of 0.66. This may be attributed to DHIL's reliance on high-quality

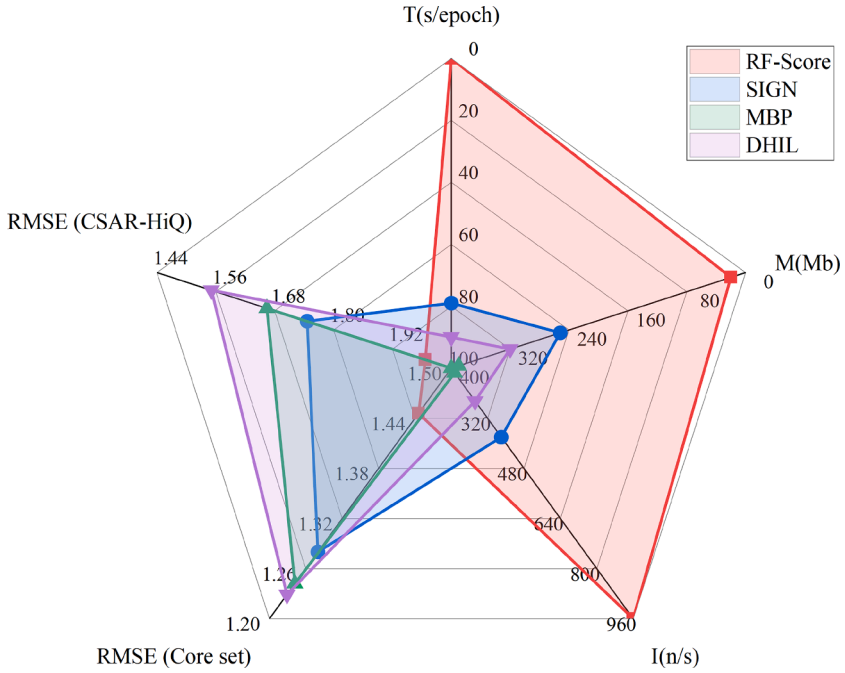


Fig. 14. Radar chart comparing the computational efficiency and predictive performance of DHIL with baseline models (RF-Score, SIGN, and MBP). The five key metrics evaluated include training time per epoch T (s/epoch), model size M (MB), inference speed I (samples/s), and root mean square error (RMSE) on both the PDBbind Core set and CSAR-HiQ dataset. Lower RMSE values indicate better predictive performance, while lower T and M values with higher I suggest better computational efficiency.

protein-ligand interaction information, while proteases and ion channels exhibit more diverse ligand binding mechanisms. Additionally, some ion channels have open ligand-binding pockets, making it more difficult to model both intramolecular and intermolecular interactions effectively. Despite these variations, DHIL outperforms traditional methods in overall screening capability across different protein categories, particularly excelling in kinase and nuclear receptor screening. These results validate DHIL's strong potential for virtual screening applications.

6. Discussion

This section aims to provide a balanced perspective on DHIL's strengths and limitations while guiding future research toward enhanced optimization and broader applicability. First, a systematic analysis of the model's computational complexity is conducted through experiments, along with a comprehensive comparison of DHIL and baseline methods in terms of computational performance to assess the trade-off between computational cost and predictive accuracy. Furthermore, the computational overhead associated with DHIL's dual-channel GNN architecture and hierarchical interaction mechanism is explored, and potential optimization strategies are proposed to improve training efficiency and inference speed. Second, the sensitivity of DHIL to the quality of input protein-ligand conformations is examined to evaluate its dependence on high-resolution crystal structures and assess its generalization capability across different conformational inputs.

6.1. Study on computational complexity

This subsection conducts a systematic investigation into the computational complexity of DHIL from both experimental and theoretical perspectives, incorporating visualized key performance indicators, analytical complexity derivations, and potential optimization strategies to comprehensively characterize the computational overhead and optimization potential of DHIL.

6.1.1. Experimental analysis of computational complexity and baseline comparison

To gain deeper insights into the trade-offs between computational cost and predictive performance, we compare DHIL's computational efficiency with existing baseline models (such as RF-Score, SIGN, and MBP). Fig. 14 visualizes multiple key indicators for this comparison, including training time (T, in seconds per epoch), model parameter size (M, in megabytes), inference speed (I, in samples per second), RMSE on the Core set, and RMSE on the CSAR-HiQ dataset. The radar chart indicates that DHIL achieves a balanced trade-off between computational cost and predictive accuracy. It attains the lowest RMSE on both the Core set (1.229) and CSAR-HiQ (1.511), demonstrating superior predictive capability. Meanwhile, its computational complexity remains within a reasonable range compared to other deep learning baseline models. In terms of training time, DHIL requires 90 s/epoch, positioning it between SIGN (79 s/epoch) and MBP (110 s/epoch). This suggests that its dual-channel hierarchical interactive learning mechanism not only enhances predictive performance but also effectively manages computational costs through structural optimization. The model's parameter size is 320 MB, slightly larger than SIGN (252 MB) but smaller than MBP (390 MB), indicating a balance between model complexity and storage requirements. For inference speed, DHIL processes 305 samples per second, significantly outperforming MBP (215 samples per second) and maintaining robust efficiency in high-precision tasks, even if it is not as fast as SIGN (420 samples per second). Overall, DHIL's optimized computational resource allocation effectively balances computational efficiency and predictive performance, making it well-suited for PLBA prediction tasks.

6.1.2. Theoretical analysis of computational complexity

Experimental analysis demonstrates that DHIL achieves a favorable balance between computational cost and prediction accuracy. However, it still incurs a certain computational burden, primarily stemming from the complexity of hierarchical interactive learning in GNN-based representation updates.

Intra-type Channel Complexity. In the DHIL model, proteins and ligands are encoded through their respective intra-type channels and a shared inter-type channel. Within the intra-type channel, the primary computational bottleneck arises from GATs, which are used to derive node representations within each molecule. For instance, the update process for intra-molecular atomic nodes has a per-layer computational complexity of:

$$O((|\mathcal{V}_L^{atom}| + |\mathcal{V}_P^{atom}|)d^2 + (|\mathcal{E}_L^{atom}| + |\mathcal{E}_P^{atom}|)d), \tag{30}$$

where this complexity reflects the computational cost associated with attention weight calculations, feature updates, and message passing. Given that intra-molecular bonding in biomolecules is typically sparse, each atom/residue has a limited number of neighbors (low average degree). Additionally, the number of ligand atoms and pocket residues is relatively small (e.g., ligand atoms on the order of tens and pocket residues fewer than a hundred). As a result, this portion of the computational overhead remains manageable.

Inter-type Channel Complexity. In the inter-type channel, the DHIL model explicitly models cross-molecular (protein-ligand) interactions by introducing a distance-based cross-molecular interaction matrix. For example, protein-ligand interactions are propagated through the interaction graph $\mathcal{G}_I^{atom}$ via cross-channel message passing. Without any filtering, the worst-case computational complexity for inter-molecular interactions is:

$$O(|\mathcal{V}_L^{atom}| \cdot |\mathcal{V}_P^{atom}| \cdot d). \tag{31}$$

It is important to emphasize that if all cross-molecular node pairs are processed indiscriminately, the computational cost will grow as the product of the ligand and protein node counts, which becomes infeasible for large proteins. To address this, DHIL attempts to reduce computational overhead by incorporating pocket residue filtering and distance thresholding. First, only protein residues within a certain range of the ligand are considered (typically, the binding site is defined as residues within 10 Å of any ligand atom). This significantly reduces the number of protein nodes from thousands to merely dozens or up to a hundred within the pocket region, effectively shrinking the effective size of $\mathcal{V}_P^{atom}$ and $\mathcal{V}_P^{sub}$, thereby reducing the computational cost of GATs and cross-channel message passing. Furthermore, as shown in Eq. (12), setting a stricter distance threshold (e.g., 6 Å or 8 Å instead of 10 Å) in the interaction matrices $\mathbf{D}^{atom}$ and $\mathbf{D}^{sub}$ further compresses the number of candidate interaction pairs, preventing unnecessary global information propagation and reducing the computational complexity of cross-molecular interactions to:

$$O(|\mathcal{V}_L^{atom}| \cdot k \cdot d), \quad k \ll |\mathcal{V}_P^{atom}|. \tag{32}$$

Hierarchical Interactive Learning Complexity. DHIL introduces bridge nodes to facilitate hierarchical cross-channel message exchange. Bridge nodes are virtual nodes that serve as information relays between intra-type channels (protein or ligand internal graphs) and inter-type channel (cross-molecular interactions), enabling distant regular nodes to communicate through shared bridge nodes. For instance, in atom-level interactive learning, atomic-level interactions are facilitated through bridge nodes $\mathcal{V}^{loc_bri}$, which mediate information exchange between ligand and protein atoms. The key steps include bridge node initialization, message passing from atomic nodes to bridge nodes, bridge node feature updates, information transfer from bridge nodes back to atomic nodes, and atomic node feature updates. The total computational complexity of this process is:

$$O(l_{atom}(|\mathcal{V}^{loc_bri}| \cdot m \cdot d + |\mathcal{V}^{loc_bri}| \cdot d^2 + |\mathcal{V}_L^{atom}| \cdot d^2)). \tag{33}$$

Compared to directly allowing all atoms to interact with all residues, the bridge node mechanism compresses interaction paths to some extent: it transforms numerous direct atom-residue communications into aggregated interactions via a small number of bridge nodes, thereby reducing communication redundancy. However, this hierarchical interaction strategy also introduces additional computational overhead, particularly from updating bridge nodes and performing multi-step interactions. If the number of bridge nodes is large or the number of interaction iterations l_{atom} or l_{sub} is high, the computational cost during inference increases accordingly. Therefore, in practical DHIL operations, a trade-off must be made between representation capability and computational efficiency. For example, we observe that performing a single interaction at both the atomic and substructure levels (i.e., two rounds of message passing) significantly improves prediction accuracy, whereas additional interaction rounds (e.g., multiple repeated atom-level interactions) yield diminishing returns while computational costs increase proportionally. The DHIL model effectively controls multi-step interactions within a manageable computational budget by selecting only the necessary hierarchical levels (atomic and substructure levels) and performing a single bridge-based message passing, thereby avoiding excessive computational overhead.

6.1.3. Potential computational complexity optimization strategies

To address the computational challenges in DHIL, potential optimization strategies such as non-bonded interaction pruning, sparse matrix operations, and attention-based bridge node sparsification can be explored to reduce the complexity of hierarchical interaction modeling.

Non-bonded Interaction Pruning. In the cross-molecular interaction matrix (such as the atom-level interaction matrix $\mathbf{D}^{atom}$), most atomic interactions exhibit only weak signals, having limited impact on final affinity and force predictions. By employing a stepwise screening strategy, redundant interactions can be pruned:

Coarse-graining based on physicochemical properties: Only atoms relevant to specific interaction types are retained. For instance, hydrogen bonding requires donors (e.g., O, N) and acceptors, while hydrophobic interactions involve only nonpolar atoms (e.g., C). Define the atomic type encoding matrix as $\mathbf{M} \in \{0,1\}^{|\mathcal{V}_L^{atom}| \times |\mathcal{V}_P^{atom}|}$:

$$\mathbf{M}_{ij} = \begin{cases} 1 & \text{if } v_{L,i}^{atom} \in \{O,N\}, v_{P,j}^{atom} \in \{O,N\} \quad \text{(Hydrogen bonding)} \\ 1 & \text{if } v_{L,i}^{atom} \in \{C\}, v_{P,j}^{atom} \in \{C\} \quad \text{(Hydrophobic interaction)} \\ 0 & \text{otherwise} \end{cases} \tag{34}$$

where the number of nonzero elements in the matrix $\mathbf{D}^{atom}$ is reduced from $\mathcal{V}_L^{atom} \times \mathcal{V}_P^{atom}$ to $\eta \times \mathcal{V}_L^{atom} \times \mathcal{V}_P^{atom}$, where η represents the proportion of elements retained after masking.

Fine-grained Pruning Based on Attention Weights: During the update process of the Inter-type Channel nodes, only significant interactions with attention weights exceeding a threshold τ are retained, thereby further refining cross-molecular interactions. For instance, the update of $\mathbf{Z}_L^{atom,(l+1)}$ is performed as follows:

$$\mathbf{Z}_{L,i}^{\text{atom},(l+1)} = \sum_{j \in \mathcal{N}_\tau(v_{L,i}^{atom})} \sigma\left(\mathbf{D}_{ij}^{atom} \mathbf{W}_4^{(l+1)} \left(\mathbf{Z}_{L,i}^{atom,(l)} \odot \mathbf{Z}_{P,j}^{atom,(l)}\right)\right), \tag{35}$$

where $\mathcal{N}_\tau(v_{L,i}^{atom}) = \{j \mid \mathbf{D}_{ij}^{atom} > \tau\}$. The threshold τ can be set based on the distribution of attention weights, such as $\tau = \mu + \alpha \cdot \sigma$, where μ and σ are the mean and standard deviation of the weights, respectively, and α is a scaling factor. The initial computational complexity per ligand atom is $O(|\mathcal{V}_P^{\text{atom}}| \cdot d)$. After optimization, it is reduced to $O(k_\tau \cdot d)$, where $k_\tau \ll |\mathcal{V}_P^{atom}|$ represents the average number of effective neighbors.

Sparse Matrix Operations. The cross-molecular interaction matrices in the DHIL model ($\mathbf{D}^{atom}$ and $\mathbf{D}^{sub}$) are predominantly sparse: only elements within a threshold distance are nonzero, accounting for a small proportion of the entire matrix. Given this sparsity, sparse matrix storage and computational optimizations can be employed. For instance, $\mathbf{D}$ can be stored using compressed sparse row (CSR) or coordinate (COO) format, recording only nonzero elements and their indices. This approach significantly reduces computational complexity in cross-channel message passing operations, such as matrix-vector multiplication. Instead of the dense matrix complexity $O(|\mathcal{V}_L| \cdot |\mathcal{V}_P|)$, the complexity is reduced to $O(E_{\text{inter}})$, which is proportional to the number of nonzero interactions. In a ligand-protein complex, if each ligand node has an average

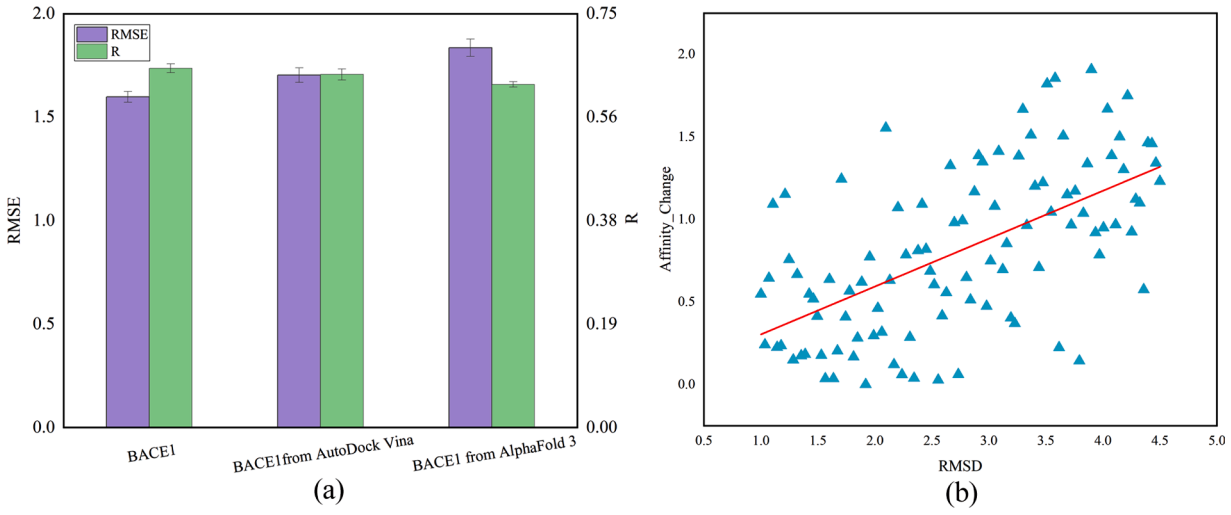


Fig. 15. Plot of PLBA predictions using different crystal structures. We utilize 20 high-resolution BACE1 crystal structures from the D3R Grand Challenge 4 (GC4) dataset as the research foundation and derives their experimental variants to assess the sensitivity of DHIL to conformational deviations and its dependence on high-quality three-dimensional complex structures. (a) RMSE and R for binding affinity predictions based on 20 BACE1 crystal structures and their structural variants, including experimentally determined BACE1 crystal structures, docking-generated structures from AutoDock Vina, and protein-ligand complex structures predicted by AlphaFold 3. The docking-generated and AlphaFold 3 structures serve as computationally derived variants of the crystal structures, exhibiting different degrees of conformational deviation. RMSE quantifies the prediction error, while R measures the correlation between predicted and experimental binding affinities. (b) Scatter plot illustrating the relationship between RMSD (root-mean-square deviation) and affinity change across 100 docking-generated structures derived from the 20 BACE1 crystal structures using AutoDock Vina. RMSD represents the structural deviation between these docking-generated complexes and their corresponding reference crystal structures, while affinity change quantifies the difference between predicted and experimental binding affinities. This analysis provides insights into how structural deviations impact the accuracy and stability of binding affinity predictions.

of k protein neighbors, the number of nonzero elements is approximately $E_{\text{inter}} \approx k \cdot |\mathcal{V}_L|$, which is significantly smaller than $|\mathcal{V}_L| \cdot |\mathcal{V}_P|$. By leveraging efficient sparse matrix multiplications, unnecessary multiplications and accumulations involving zero elements can be avoided.

Dynamic Pruning of Bridge Nodes. In hierarchical interactive learning, bridge nodes serve as relay hubs for cross-channel information transfer, but may contain redundant connections. To reduce computational overhead, a dynamic pruning strategy based on attention weights is proposed: first, define the importance score of a bridge node $v_{L,i}^{\text{loc_bri}}$ as the maximum value of its attention gating coefficients, i.e., $s_i = \max\left(\mathbf{g}_{L,i}^{\text{loc}^{(l)}}\right)$, where $\mathbf{g}_{L,i}^{\text{loc}^{(l)}} \in \mathbb{R}^d$ represents the attention gating coefficients of the bridge node at layer l; second, set a global threshold τ_{prune} to retain only bridge nodes with importance scores exceeding the threshold, i.e., $\mathcal{V}_{\text{active}}^{\text{loc_bri}} = \left\{ v_{L,i}^{\text{loc_bri}} \mid s_i > \tau_{\text{prune}} \right\}$, where the retention ratio β satisfies $\beta = |\mathcal{V}_{\text{active}}^{\text{loc_bri}}|/|\mathcal{V}^{\text{loc_bri}}|$; through pruning, the computational complexity of bridge node-related operations (feature aggregation and updates) is reduced from $O\left(l_{\text{atom}} \cdot |\mathcal{V}^{\text{loc_bri}}| \cdot (m \cdot d + d^2)\right)$ to $O\left(l_{\text{atom}} \cdot \beta \cdot |\mathcal{V}^{\text{loc_bri}}| \cdot (m \cdot d + d^2)\right)$, where m is the number of atoms associated with each bridge node and d is the feature dimension; additionally, dynamically adjust the threshold based on the distribution of attention weights to ensure the retention of critical interactions, i.e., $\tau_{\text{prune}} = \mu_s + \alpha \cdot \sigma_s$, where μ_s and σ_s are the mean and standard deviation of the importance scores within a batch, and α is a control parameter.

6.2. Conformational sensitivity assessment of crystal structures

In this subsection, 20 high-resolution BACE1 crystal structures from the D3R Grand Challenge 4 (GC4) (Parks et al., 2020) dataset are selected as the research objects, along with their experimental variants, to evaluate the sensitivity of DHIL to conformational deviations and its dependence on high-quality three-dimensional complex structures. As a key benchmark in computational chemistry, GC4 provides co-crystal structures of BACE1 inhibitors along with binding affinity data, including 20 complexes with high-resolution crystal structures, while others require docking or modeling approaches to predict binding conformations. This characteristic makes GC4 an ideal dataset for investigating the impact of protein-ligand complex structure quality on binding affinity prediction. By analyzing the geometric deviations (RMSD) between the 20 crystal structures and their experimental variants, we assess the extent to which DHIL relies on the accuracy of structural inputs and evaluate its applicability to docking-predicted structures (via AutoDock Vina (Trott & Olson, 2010)) or AlphaFold-predicted complexes (via AlphaFold 3 (Abramson et al., 2024)).

6.2.1. Impact of different conformational inputs on DHIL prediction performance

Fig. 15(a) presents the variations in RMSE and R across different conformational inputs, including BACE1 crystal structures, docking-generated structures from AutoDock Vina, and protein-ligand complex structures predicted by AlphaFold. The crystal structures exhibit the best prediction performance, characterized by the lowest RMSE and the highest R, indicating that DHIL can effectively learn protein-ligand interactions from high-quality experimental structures, ensuring reliable binding affinity predictions. In contrast, the RMSE of docking-generated structures increases while R decreases, suggesting that deviations in binding modes from the crystal structures may lead to the loss of key interactions such as hydrogen bonds, hydrophobic contacts, or salt bridges. Furthermore, docking methods typically assume a rigid protein model, overlooking the flexibility of the binding pocket, which exacerbates structural deviations and reduces prediction accuracy. For AlphaFold-predicted structures, RMSE further increases while R declines, likely due to discrepancies in binding pocket geometry, including inconsistencies in global folding relative to experimental structures and variations in side-chain orientations. These deviations can significantly alter ligand binding modes, impairing the model's ability to extract meaningful interaction features.

6.2.2. Conformational deviations have a significant impact on DHIL

Fig. 15(b) illustrates the relationship between RMSD (conformational deviation) and binding affinity prediction error, where larger RMSD values correspond to greater deviations in predicted affinity,

reflecting the sensitivity of DHIL to input structural variations. When RMSD < 2.0 Å, affinity predictions remain relatively stable, indicating that DHIL can accommodate minor conformational changes while maintaining high predictive accuracy. However, when RMSD exceeds 3.0 Å, prediction errors increase significantly, suggesting that beyond a certain threshold, DHIL struggles to correctly capture key interactions, leading to unstable and unreliable affinity predictions. This trend is likely driven by substantial shifts in binding modes, where higher RMSD values result in ligand misalignment within the binding pocket, preventing DHIL from accurately extracting key interaction features and ultimately reducing prediction accuracy. The fitted linear trend confirms a positive correlation between RMSD and affinity prediction error, demonstrating that greater structural deviations contribute to increased prediction uncertainty. This finding further supports the conclusion drawn from Fig. 15(a), where DHIL performs best with high-quality input structures (crystal structures), while errors in docking-generated and AlphaFold-predicted structures may lead to incorrect modeling of binding modes, thereby reducing prediction accuracy.

6.2.3. Optimization strategies for conformational variations

The above experimental results indicate that DHIL is highly sensitive to conformational variations and relies on high-quality crystal structure inputs. When applied to docking-generated structures and AlphaFold-predicted complexes, the model exhibits significant prediction errors, highlighting its limited adaptability to substantial conformational deviations. Future research can focus on the following directions: (1) Enhancing the model's robustness to conformational variations through contrastive learning and self-supervised learning approaches, enabling it to learn protein-ligand interaction features across diverse conformations and reducing its dependence on high-resolution crystal structures. Integrating protein language models can leverage large-scale protein sequence pretraining to improve the model's ability to capture functional and structural characteristics, thereby enhancing its adaptability to low-quality or predicted conformational data. (2) Improving rigid docking strategies and incorporating flexible modeling by addressing the limitations of conventional docking methods that assume rigid protein structures. This can be achieved by introducing molecular dynamics simulations to explore the conformational flexibility of binding pockets and leveraging diffusion models and other deep learning approaches to generate realistic binding pocket conformations. Such improvements can lead to more accurate ligand binding mode predictions while mitigating biases introduced by rigid docking assumptions. (3) Implementing multi-task learning strategies to enable simultaneous prediction of binding affinity, conformational optimization, and binding site identification, thereby improving the model's generalization ability. By integrating information across different hierarchical levels, the model can enhance its adaptability to diverse environments and exhibit greater stability and practical utility in dynamic protein-ligand interaction studies. These optimizations will contribute to advancing research in more complex biological systems, allowing the model to maintain high prediction accuracy across different conformational inputs and further improving its applicability in real-world scenarios.

7. Conclusion

In this study, we propose a Dual-channel Hierarchical Interactive Learning (DHIL) framework to improve the accuracy and interpretability of protein-ligand binding affinity prediction. DHIL introduces a dual-channel encoding strategy to independently learn intramolecular (covalent) and intermolecular (non-covalent) interactions, and further employs a hierarchical interactive learning mechanism that enables bidirectional information exchange through local and global bridge nodes. This architecture effectively captures the hierarchical synergistic effects between atomic-level and substructure-level interactions, aligning with the multiscale nature of biochemical binding processes. Extensive experiments on multiple benchmark datasets demonstrate that DHIL consistently outperforms state-of-the-art baselines in terms of predictive accuracy, robustness, and generalization ability. Moreover, ablation studies and visualizations validate the benefits of hierarchical interaction modeling and the bridge-based feature fusion strategy.

Although DHIL demonstrates strong predictive performance, it still faces two notable limitations. The hierarchical graph construction and bridge-based interaction mechanisms introduce considerable computational overhead, while the model remains highly sensitive to the accuracy of input three-dimensional conformations. These factors may hinder its scalability and robustness in practical applications involving large-scale data or predicted structural inputs. To address these challenges, future research will focus on improving computational efficiency through sparsity-aware modeling and optimized bridge interaction strategies. In addition, integrating contrastive learning, pretrained protein representations, and flexible binding site modeling is expected to enhance the model's adaptability to structural variability, thereby broadening the applicability of DHIL across diverse drug discovery scenarios.

CRediT authorship contribution statement

Zheyu Wu: Methodology, Data curation, Writing – original draft, Investigation, Writing – review & editing, Resources, Formal analysis, Conceptualization; **Huifang Ma:** Resources, Conceptualization, Supervision, Funding acquisition, Project administration; **Bin Deng:** Validation, Conceptualization; **Zhixin Li:** Validation, Funding acquisition, Supervision; **Liang Chang:** Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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